September 4, 2025

Dresner Advisory Services, LLC

2025 Edition

ModelOps Market Study

Wisdom of Crowds® Series

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Disclaimer:

This report is for informational purposes only. You should make vendor and product selections based on multiple information sources, face-to-face meetings, customer reference checking, product demonstrations, and proof-of-concept applications.

The information contained in this Wisdom of Crowds® Market Study report is a summary of the opinions expressed in the online responses of individuals that chose to respond to our online questionnaire and does not represent a scientific sampling of any kind. Dresner Advisory Services, LLC shall not be liable for the content of this report, the study results, or for any damages incurred or alleged to be incurred by any of the companies included in the report as a result of the report's content.

Definitions

Business Intelligence Defined

Business intelligence (BI) is “knowledge gained through the access and analysis of business information.”

Business intelligence tools and technologies include query and reporting, online analytical processing (OLAP), data mining and advanced analytics, end-user tools for ad hoc query and analysis, and dashboards for performance monitoring.

Howard Dresner, *The Performance Management Revolution: Business Results Through Insight and Action* (John Wiley & Sons, 2007)

ModelOps Defined

Model operations (ModelOps) refers to the set of practices, processes, and technologies used to manage and operationalize machine learning (ML) models throughout their life cycle, from development and deployment to monitoring and maintenance. This approach ensures that models are effectively integrated into production environments and deliver consistent, reliable results. ModelOps includes, but is not limited to, machine learning (ML) and artificial intelligence (AI) models, including large language models (LLM) as well as less-complex analytical and decision-intelligence models.

Introduction

This fourth annual edition of our ModelOps Market Study coincides with the eighteenth anniversary of Dresner Advisory Services.

As organizations expand their use of AI, data science, and machine learning, ModelOps has become a critical discipline for managing models at scale.

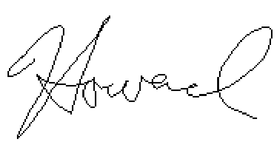
Our research shows continued growth in the number and variety of models in production, but also persistent challenges with visibility, oversight, and ownership. Increasingly, responsibility for ModelOps is shifting toward centralized functions to strengthen governance and infrastructure.

The growing influence of agentic AI underscores the urgency of robust ModelOps practices, extending their importance beyond AI and machine learning to all analytical models. As adoption deepens, ModelOps will be central to ensuring adaptability, accountability, and value from models in production.

As we continue to grow and evolve, we extend our sincere appreciation to our clients and community for their support over the past 18 years. Your participation and engagement have been instrumental in shaping our research and driving its relevance and impact.

We thank you for your continued trust in Dresner Advisory Services as a reliable source of knowledge and expertise.

Best,



Chief Research Officer
Dresner Advisory Services

Contents

Benefits of the Study

This Dresner Advisory Services ModelOps Market Study provides a wealth of information and analysis, offering value to both consumers and producers of business intelligence technology and services.

Consumer Guide

As an objective source of industry research, the Dresner Advisory Services ModelOps Market Study helps consumers to understand how their peers leverage and invest in ModelOps technologies.

Using our unique vendor performance measurement system, users glean key insights into BI software supplier performance, which enables:

- Comparisons of current vendor performance to industry norms
- Identification and selection of new vendors.

Supplier Tool

Vendor licensees use the Dresner Advisory Services ModelOps Market Study in several important ways:

External Awareness

- Build awareness for business intelligence markets and supplier brands, citing ModelOps Market Study trends and vendor performance
- Gain lead and demand generation for supplier offerings through association with Dresner Advisory Services ModelOps Market Study brand, findings, webinars, etc.

Internal Planning

- Refine internal product plans and align with market priorities and realities as identified in the Dresner Advisory Services ModelOps Market Study
- Better understand customer priorities, concerns, and issues
- Identify competitive pressures and opportunities.

About Howard Dresner and Dresner Advisory Services

The Dresner Advisory Services ModelOps Market Study was conceived, designed, and executed by Dresner Advisory Services, LLC—an independent advisory firm—and Howard Dresner, its president, founder and chief research officer.

Howard Dresner is one of the foremost thought leaders in business intelligence and performance management, having coined the term "business intelligence" in 1989. He has published two books on the subject, *The Performance Management Revolution – Business Results through Insight and Action* (John Wiley & Sons, Nov. 2007) and *Profiles in Performance – Business Intelligence Journeys and the Roadmap for Change* (John Wiley & Sons, Nov. 2009). He lectures at forums around the world and is often cited by the business and trade press.

Prior to Dresner Advisory Services, Howard served as chief strategy officer at Hyperion Solutions and was a research fellow at Gartner, where he led its business intelligence research practice for 13 years.

Howard has conducted and directed numerous in-depth primary research studies over the past two decades and is an expert in analyzing these markets.

Through the Wisdom of Crowds® Business Intelligence market research reports, we engage with a global community to redefine how research is created and shared. Other research reports include:

- Wisdom of Crowds® Flagship BI Market Study
- AI, Data Science, and Machine Learning
- Active Data Architecture®
- Agentic AI
- Analytical Data Infrastructure
- Analytical Platforms
- Data and Analytics Governance
- Data Engineering
- Embedded Business Intelligence
- Generative AI
- Semantic Layer

You can find more information about Dresner Advisory Services at www.dresneradvisory.com.

About Bill Hostmann

Bill Hostmann is a research fellow with Dresner Advisory. His area of focus includes

trends in analytic data infrastructures (ADI)—integrating and managing the information and information models used by BI, advanced analytics, and CPM/PM applications.

Bill has more than 20 years of product management experience at the intersection of business intelligence/analytics and data analytics infrastructure, including positions in product and general management at Gemstone Systems, Informix, and Informatica. In addition, Bill spent 14 years as a research analyst at Gartner, including several years as a VP and distinguished analyst for BI/analytics.

Bill's academic education includes BSEE, MSCS&EE, and MBA degrees.

The Dresner Team

About Elizabeth Espinoza

Elizabeth is director of analytics at Dresner Advisory and is responsible for the data preparation, analysis, and creation of charts for Dresner Advisory reports.

About Sherry Fairchok

Sherry is senior editor at Dresner Advisory, ensuring the quality and consistency of all research publications.

About Danielle Guinebertiere

Danielle is vice president of client services at Dresner Advisory. She supports the ongoing research process through her work with executives at companies included in Dresner market reports.

About Michelle Whitson-Lorenzi

Michelle is director of research operations and is responsible for managing software company survey activity and our internal market research data.

Survey Method and Data Collection

As with all our Wisdom of Crowds® market studies, we constructed a survey instrument to collect data and used social media and crowdsourcing techniques to recruit participants.

Data Quality

We carefully scrutinized and verified all respondent entries to ensure that only qualified participants were included in the study.

Executive Summary

Executive Summary

- **Model Proliferation and Visibility:**
  - Most organizations (39%) have 25 or fewer AI, data science, and ML models in production. However, a significant portion (51%) of respondents don't know how many models they have in production, suggesting a general lack of visibility or tracking.
  - The financial services and healthcare industries report using the largest number of models.
  - Model usage generally increases with organization size, with the largest organizations having over 250 models in production.
  - Organizations with agentic AI in production today have a large number and wide variety of models in use, indicating that advanced AI adoption correlates with increased model awareness and management.

- **Shifting Responsibility for ModelOps Oversight:**
  - According to 40% of respondents, the IT department has become the party primarily responsible for ModelOps, significantly surpassing the data science team (24%) in 2025. This is a substantial shift, possibly due to increased model scale requiring dedicated infrastructure management, a strategic decision to centralize operations for better governance, or data science teams focusing on advanced modeling.
  - Industry significantly shapes ownership models:
    - Business services and healthcare show strong IT leadership (as indicated by over 40% of those industry respondents).
    - Data science teams are most prominent in retail & wholesale (50%) and healthcare (44%), as well as education (23.53%).
    - Financial services demonstrate a mixed approach, with both IT and data science teams sharing responsibility. These variations likely reflect industry-specific risk profiles, regulatory demands, and levels of AI adoption maturity.

- **Varied Model Refresh Frequencies:**
  - The most common model refresh frequency across all regions is monthly (35% of respondents).
  - Model refresh frequency is shortest (daily) among respondents in the technology, healthcare, and consumer services industries.
  - Industry-specific patterns are clear:
    - Consumer services dominates daily refreshes (67%), driven by the need for instant personalization.

- Sales & marketing also leads with frequent daily (25%) and weekly (38%) updates for real-time personalization.
- Financial services distributes evenly across daily to quarterly, but maintain a significant 25% annual share, likely for audit purposes.
- Healthcare spreads evenly across all frequencies, including biannual (24%) and annual (24%), suggesting adherence to regulatory compliance cycles.
- Government and retail & wholesale show a strong annual tilt (33%-30%), while education's high annual rate (47%) signals long-term curriculum or policy modeling.

◦ For organizations in production with agentic AI workflows, the model refresh schedule is relatively conservative, focusing on stability and risk mitigation (25% weekly/monthly, 13.9% daily, 22% quarterly). Teams experimenting with agentic AI push for more frequent iteration (20% daily refreshes) alongside a high annual cadence (22%) for rapid prototyping and continuous learning.

• ModelOps Feature Priorities:

◦ Respondents consistently prioritize model version control, monitoring and alerting, model life cycle management, model performance monitoring, and continuous integration, delivery, and training (CI/CD/CT) as the most important capabilities. End-to-end model governance, including life cycle management (45% of respondents), lineage/history (43%), and version control (42%), also rank highly, indicating a need for traceability and reproducibility.

◦ Priorities vary significantly by industry:

- Technology and manufacturing consistently rate highest on core ModelOps capabilities like model life-cycle management (3.5–3.44) and CI/CD/CT (≈ 3.53), reflecting mature ModelOps pipelines.
- Consumer services emphasizes monitoring and alerting (3.42) and performance monitoring (3.62), highlighting a focus on real-time model health for customer-facing applications.
- Financial services prioritizes model version control (3.46) but scores lower on drift management (2.92) and external metrics (2.84), suggesting tighter governance but less automation.
- Healthcare values model lineage (3.06) and collaborative tools (3.19) but lags in feature store adoption (2.82).
- The government sector shows the highest monitoring score (3.71) yet very low feature-store use (1.71), suggesting a strong need for oversight but potentially underdeveloped data infrastructure.

• Vendor Support and Gaps:

◦ Many vendors already support essential ModelOps capabilities such as model life cycle management, monitoring and alerting, and model version control.

◦ However, there are gaps in vendor support for advanced capabilities like live A/B testing frameworks, automated remediation workflows, and feature stores. These advanced capabilities, crucial for dynamic AI systems, are often still largely unplanned by vendors.

• The Rise of ModelOps and Agentic AI Evolution:

◦ While ModelOps currently ranks 50th out of 59 strategic initiatives for business intelligence, it is anticipated to rise in importance as organizations develop and manage more models.

◦ ModelOps is broadly defined to include machine learning (ML) and artificial intelligence (AI) models, including large language models (LLMs), inherently covering agentic AI models. But organizations must apply the concepts to less-complex analytical models as well.

◦ The increasing adoption and sophistication of AI, including agentic AI, will undoubtedly drive the evolution and importance of ModelOps, creating a demand for more robust, tailored, and potentially industry-specific ModelOps solutions. This will necessitate sophisticated capabilities for continuous learning, adaptation, enhanced governance, and traceability for dynamic AI systems.

Study Demographics

Study Demographics

Participants provide a cross-section of data across geographies, functions, organization sizes, and vertical industries. We constructed cross-tab analyses using these demographics to identify and illustrate important industry trends.

Geography

Survey respondents represent a span of geographies. North America, which includes the United States, Canada, and Puerto Rico, accounts for about 59% of our survey base. EMEA respondents are the next-largest group with about 19%, followed by Asia Pacific and Latin America (fig. 1).

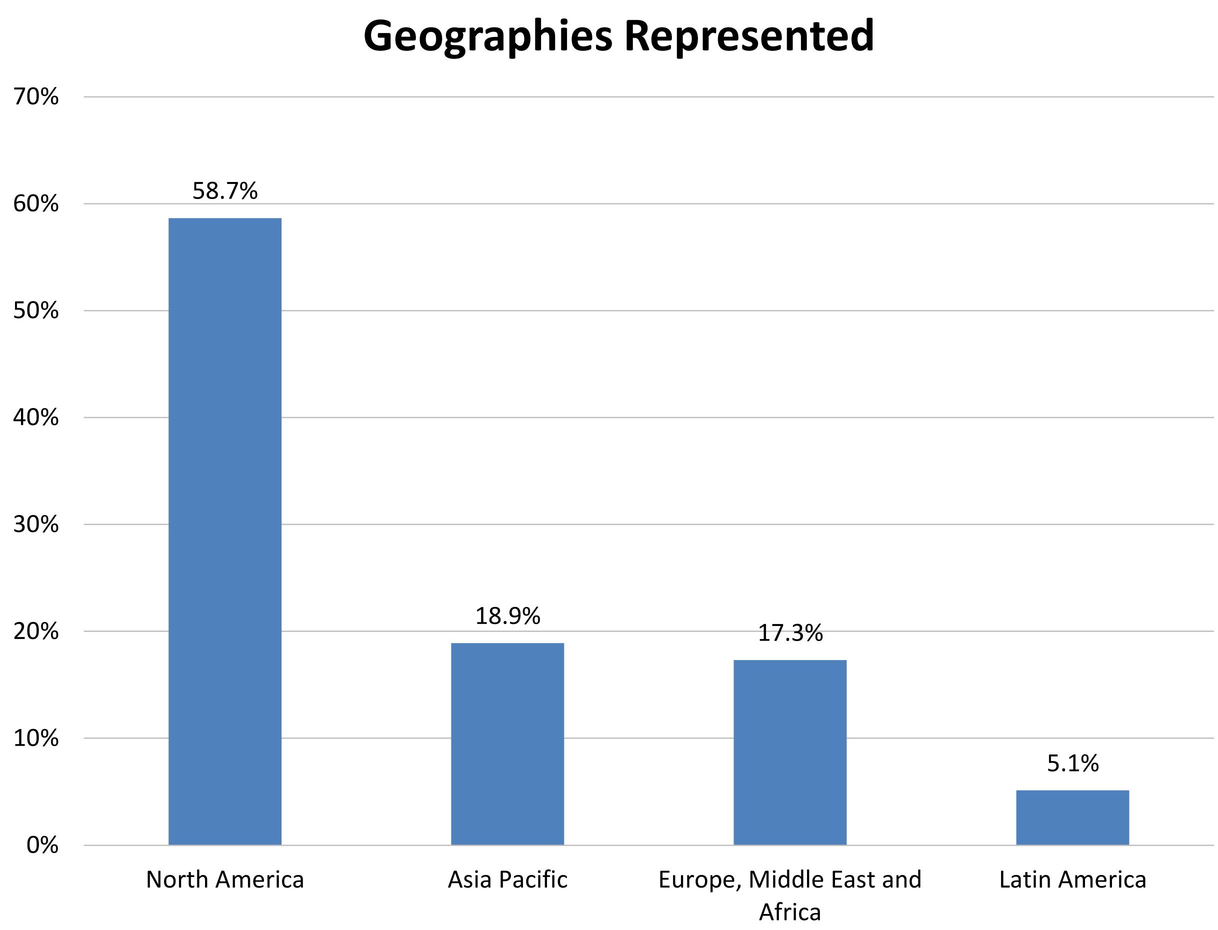


Figure 1 – Geographies represented

Functions

In our 2025 study, IT comprises the largest group by percentage (35%) in our sample base (fig. 2). Finance (16%), BICC (15%), and executive management (15%) and are the next most represented functions. Tabulating results by function enables us to compare the plans and priorities of different business functions within organizations.

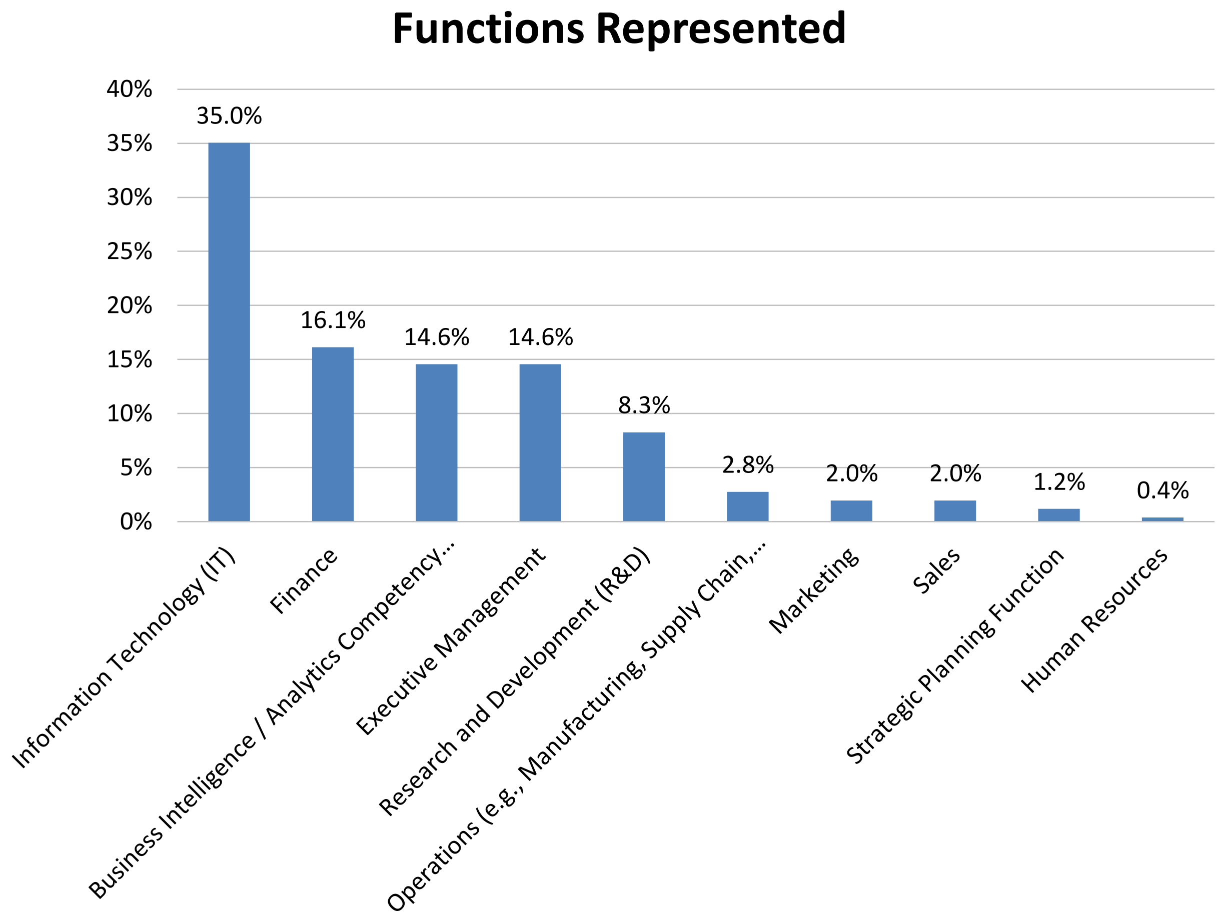


Figure 2 – Functions represented

Vertical Industries

Our study base includes a cross-section of vertical industries. Technology, manufacturing, and business services are the most represented, followed by financial services and healthcare (fig. 3).

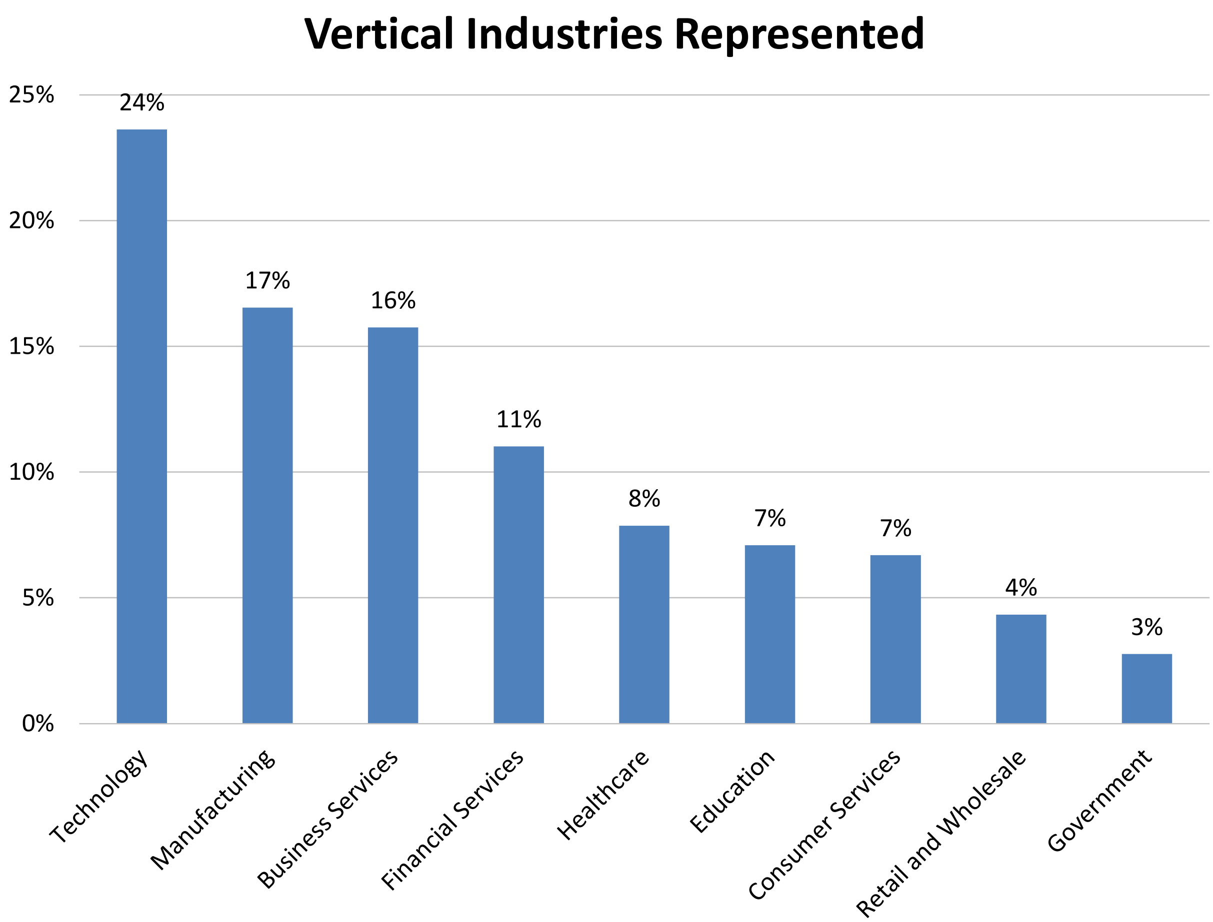


Figure 3 – Vertical industries represented

Organization Size

In our 2025 study, our survey base reflects a mix of small, midsize, and large organizations (fig. 4). Small organizations (1-100 employees) account for 25% of the sample; midsize organizations (101-1,000 employees) account for 27%; and the remaining 48% are spread across large organizations with more than 1,000 employees. Segmenting respondents by organization size helps us identify differences in behavior, attitudes, and planning often related to headcount.

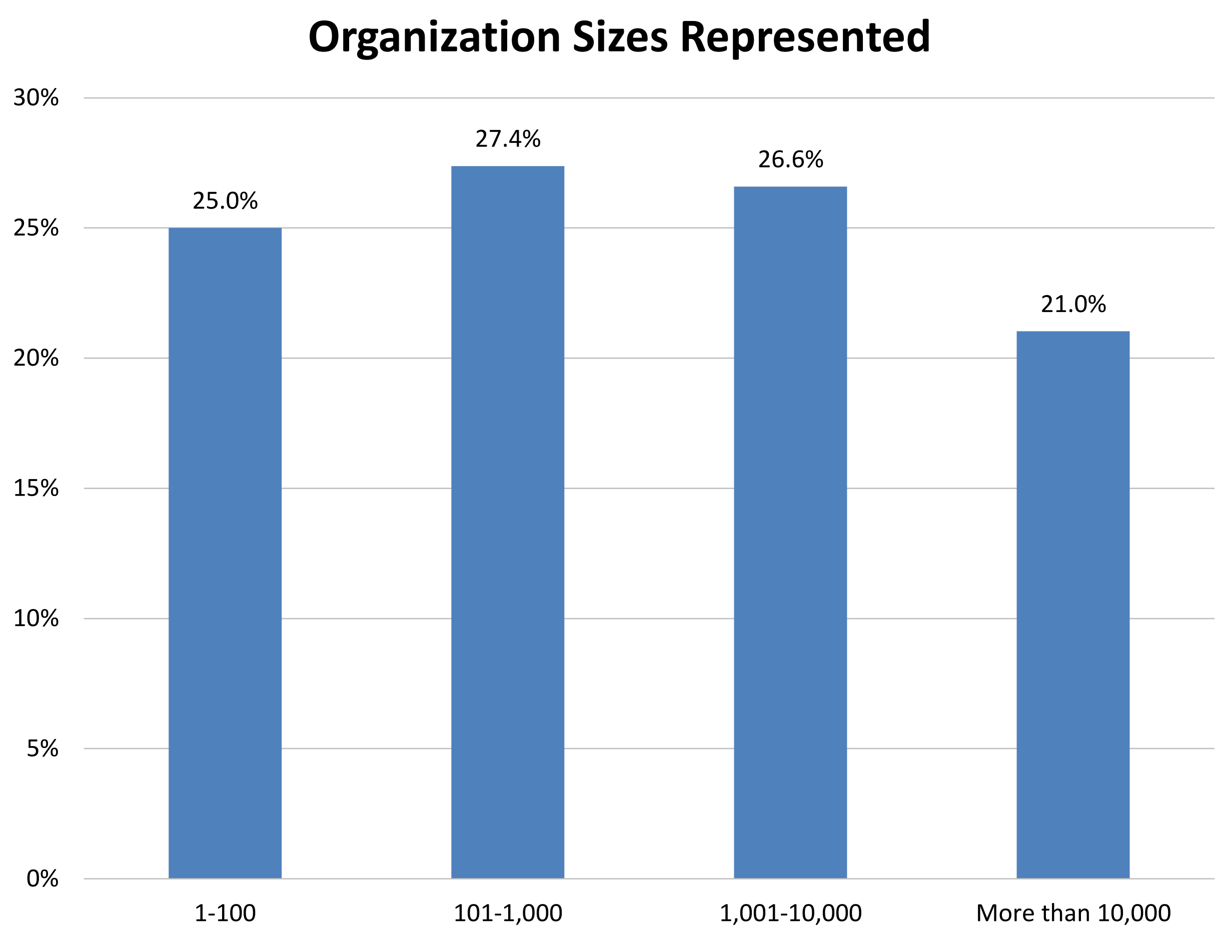


Figure 4 – Organization sizes represented

Analysis and Trends

Analysis and Trends: ModelOps

Technology Trends

ModelOps ranks 54 out of 65 initiatives strategic for BI. In comparison, AI, data science and Machine Learning rank 26 in priority. In comparison, AI, data science and machine learning rank 26th in priority. We anticipate that ModelOps will rise in importance as organizations develop and manage more models and users become aware of the value of ModelOps workflows and platforms.

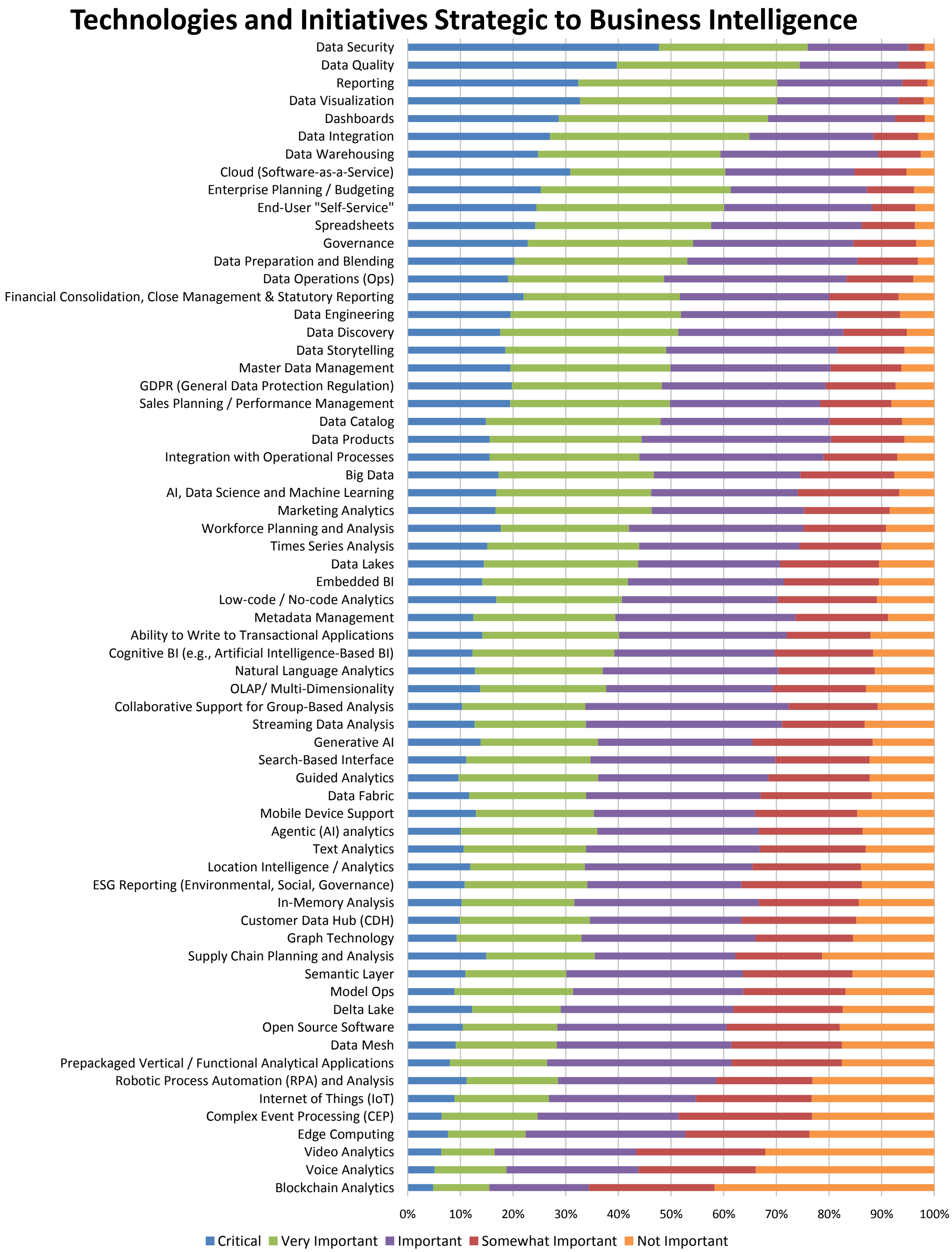


Figure 5 – Technologies and initiatives strategic to business intelligence

Number of AI, Data Science and Machine Learning Models in Use

Since 2018, we have asked survey respondents about the number of AI, data science, and machine learning models in production within their organizations. Throughout this time span, the great majority of those responding estimated their number of models in use at 25 or less or said they “don’t know” the number of models in use. In 2025, a small percentage (34%) of organizations reported having between one and 25 models, while a majority (54%) did not know the number of DS/ML models they had in use. The percentages of organizations with 26-50, 51-100, and 101-250 models remained relatively low (6%, 4%, and 2%, respectively).

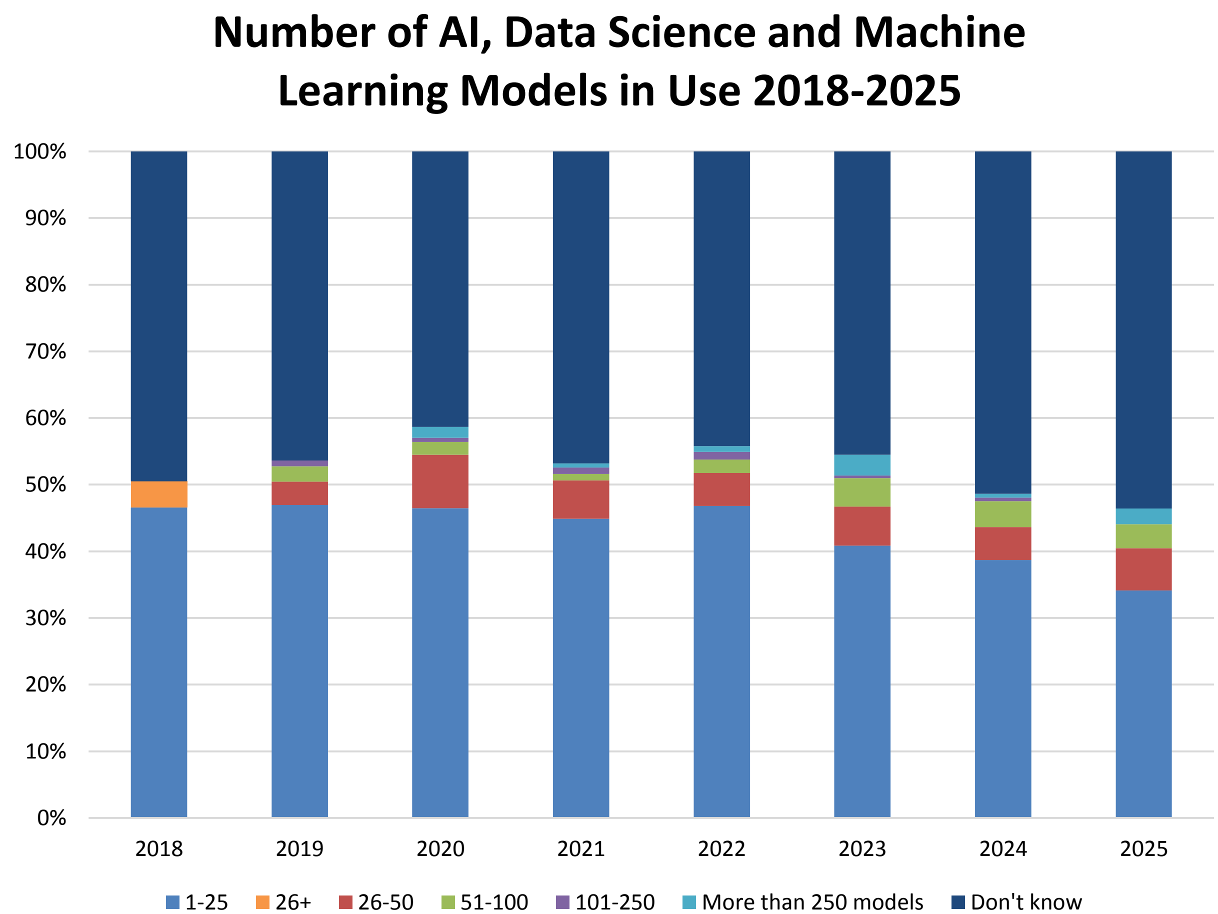


Figure 6 – Number of AI, data science and machine learning models in use 2018-2025

The following charts and analysis represent the distribution of the number of AI, data science, and machine learning models used within different geographic regions. Across all regions (Latin America, Europe/Middle East/Africa, North America, Asia Pacific), the largest portion of respondents indicated they "don't know" how many models their organization uses. This significant finding suggests a lack of visibility or tracking in many organizations. Asia Pacific respondents indicated the highest number of models in use while Latin America respondents indicated the lowest use.

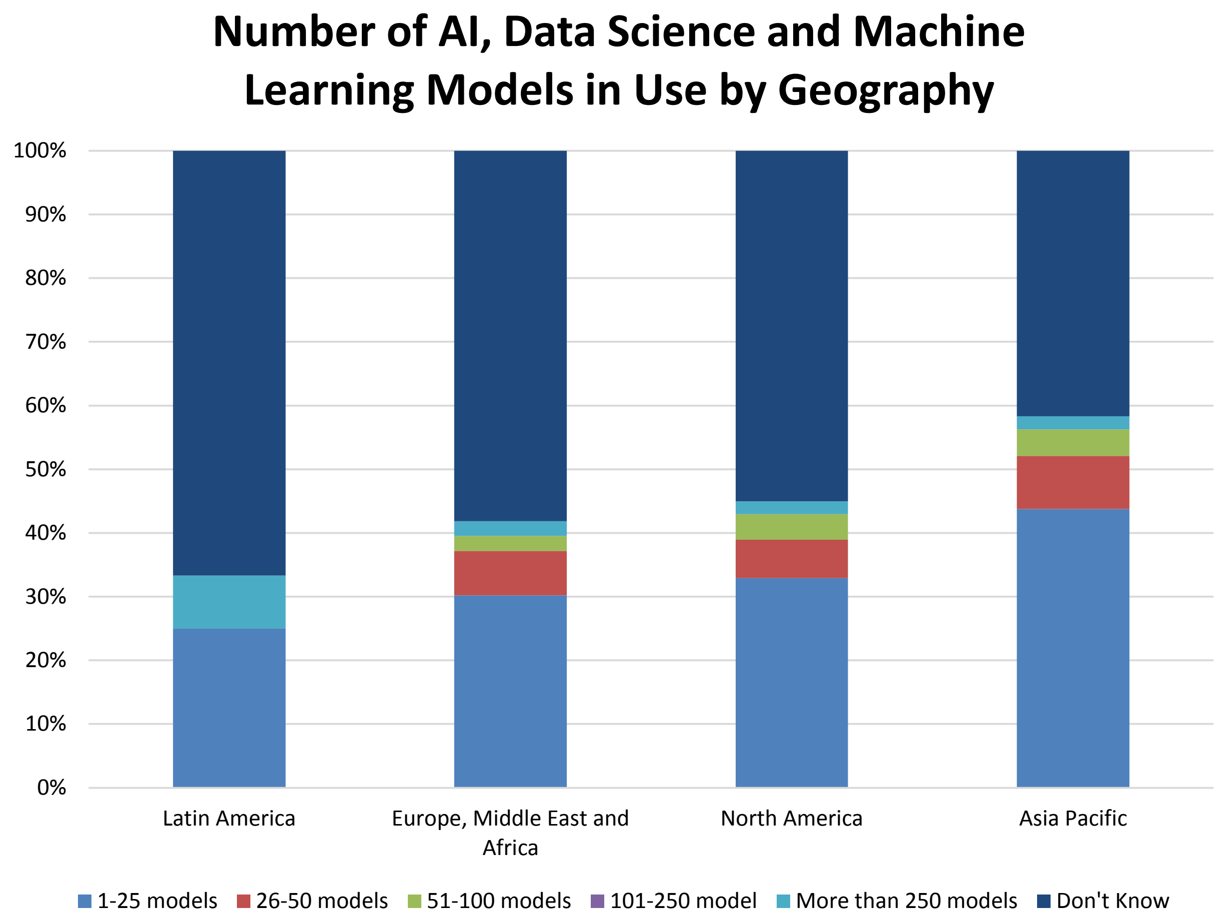


Figure 7 – Number of AI, data science, and machine learning models in use by geography

The survey data underscores the diverse landscape of model usage across different functions, with varied numbers of models in use. This suggests that some organizations lack clarity on the exact number of models being used, potentially highlighting a need for improved visibility and tracking of model utilization, effectiveness, and value. Understanding these pattern variations can aid organizations in aligning their resources and strategies to effectively leverage AI, data science, and machine learning capabilities within specific functional areas and across functions.

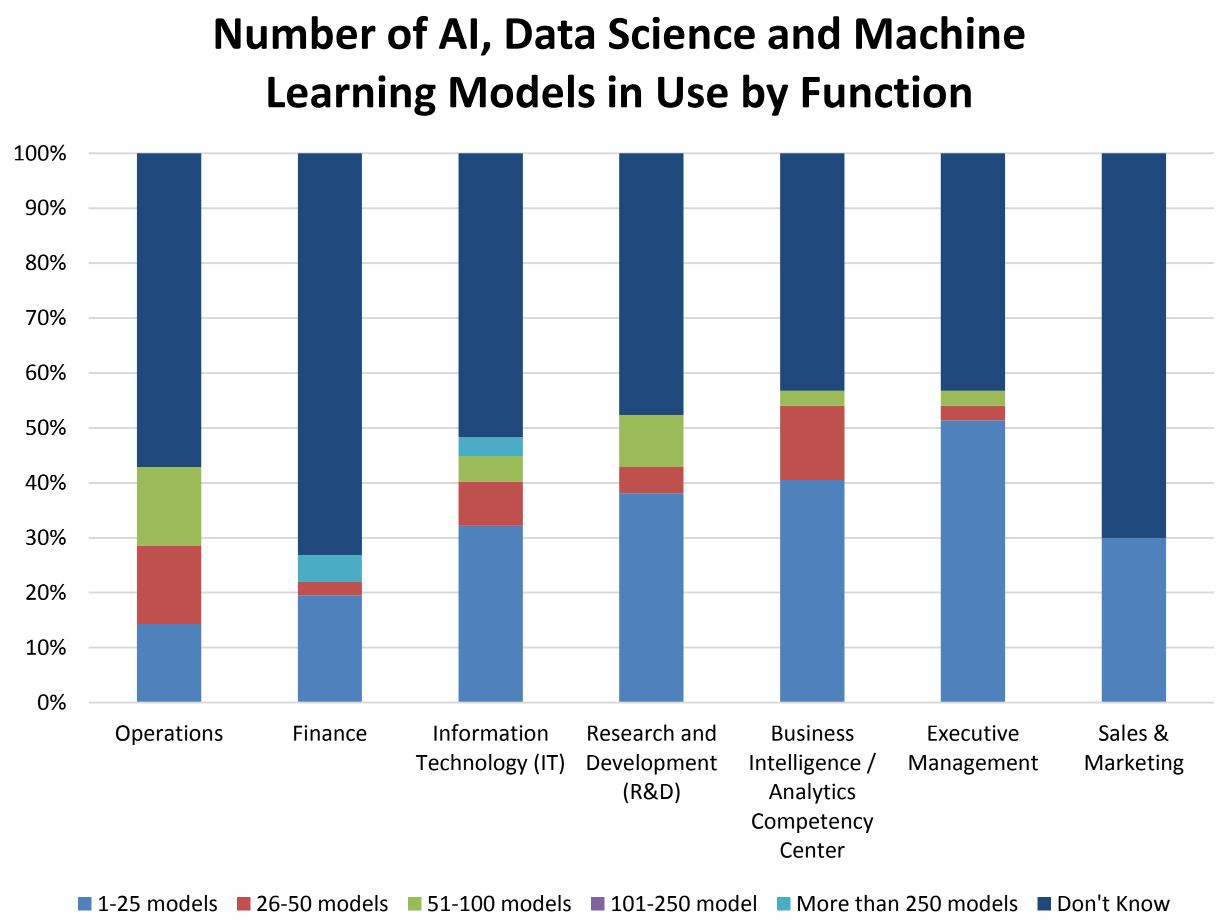


Figure 8 – Number of AI, data science, and machine learning models in use by function

The data highlights variations in the distribution of models used across different industries. Some industries have a higher proportion of organizations with unknown model counts while others show a higher usage of specific model ranges. Financial services and healthcare are the industries reporting use of the largest number of models.

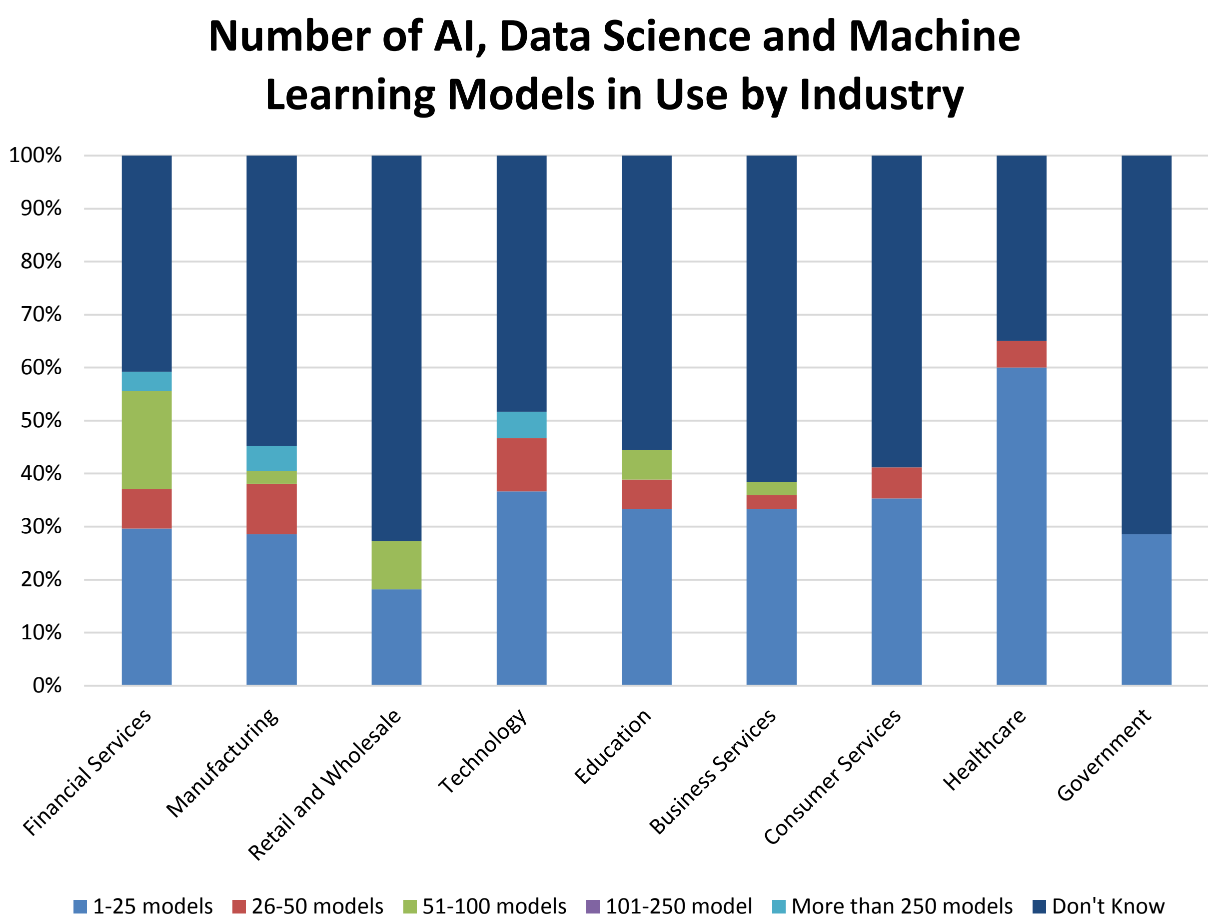


Figure 9 – Number of AI, data science and machine learning models in use by industry

The use and number of AI, data science, and machine learning models increase with organization size, with 11% of the largest organizations indicating that they have more than 250 models in use.

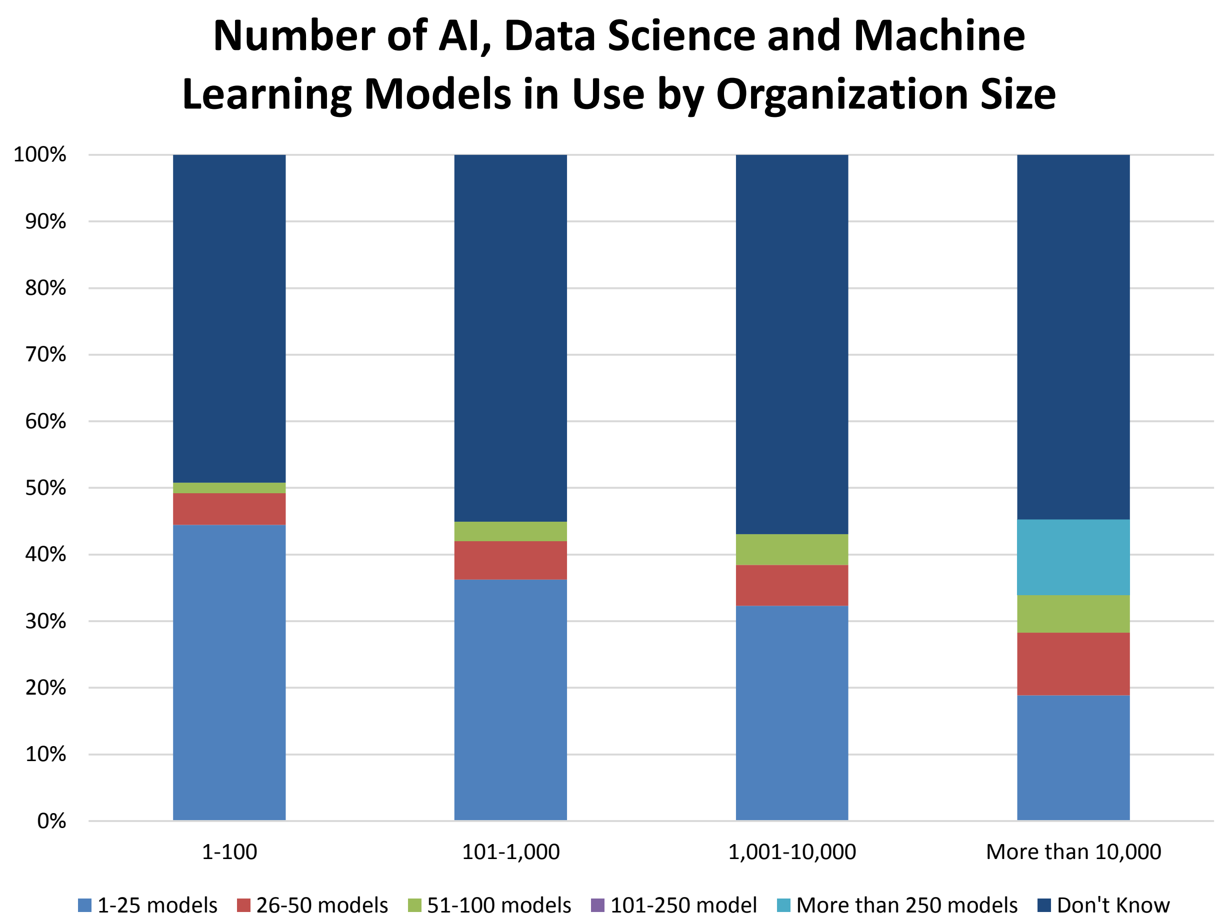


Figure 10 – Number of AI, data science and machine learning models in use by organization size

When we ask survey respondents to indicate the level of agentic AI adoption, we can see that those who have agentic AI in production today use a large number and variety of models. Those with no plans have the highest number of "don't know" responses.

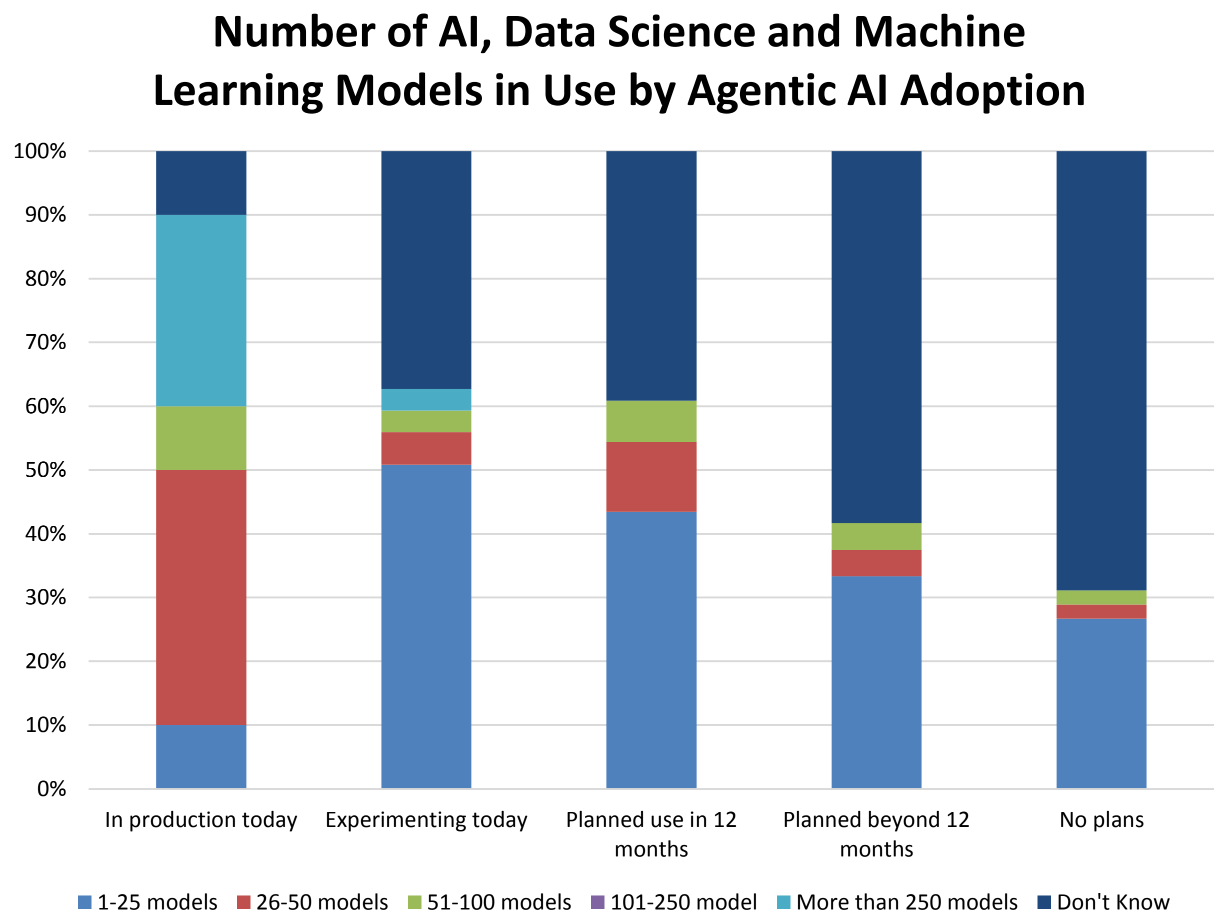


Figure 11 – Number of AI, data science, and machine learning models in use by agentic AI adoption

Responsibility for Data Science and Machine Learning Models

We asked respondents, "Who is responsible for monitoring/managing AI, ML, and predictive models (i.e., ModelOps)?" and provided a mix of choices for oversight authority by department. Beginning in 2020, we offered the option of selecting the data science team, which respondents identify as the top or most ascendant department alongside IT. This year shows a major change. The IT department takes over as the primary responsible party (40%), significantly surpassing the data science team (24%). This represents a substantial shift in responsibility and warrants further investigation to understand its causes. The dramatic shift to IT in 2025 could be due to several factors: a) the increased scale of models requiring dedicated infrastructure management; b) a strategic decision to centralize model operations for better governance and control; or c) the data science team being freed up for more advanced modeling work, while IT handles the operational aspects.

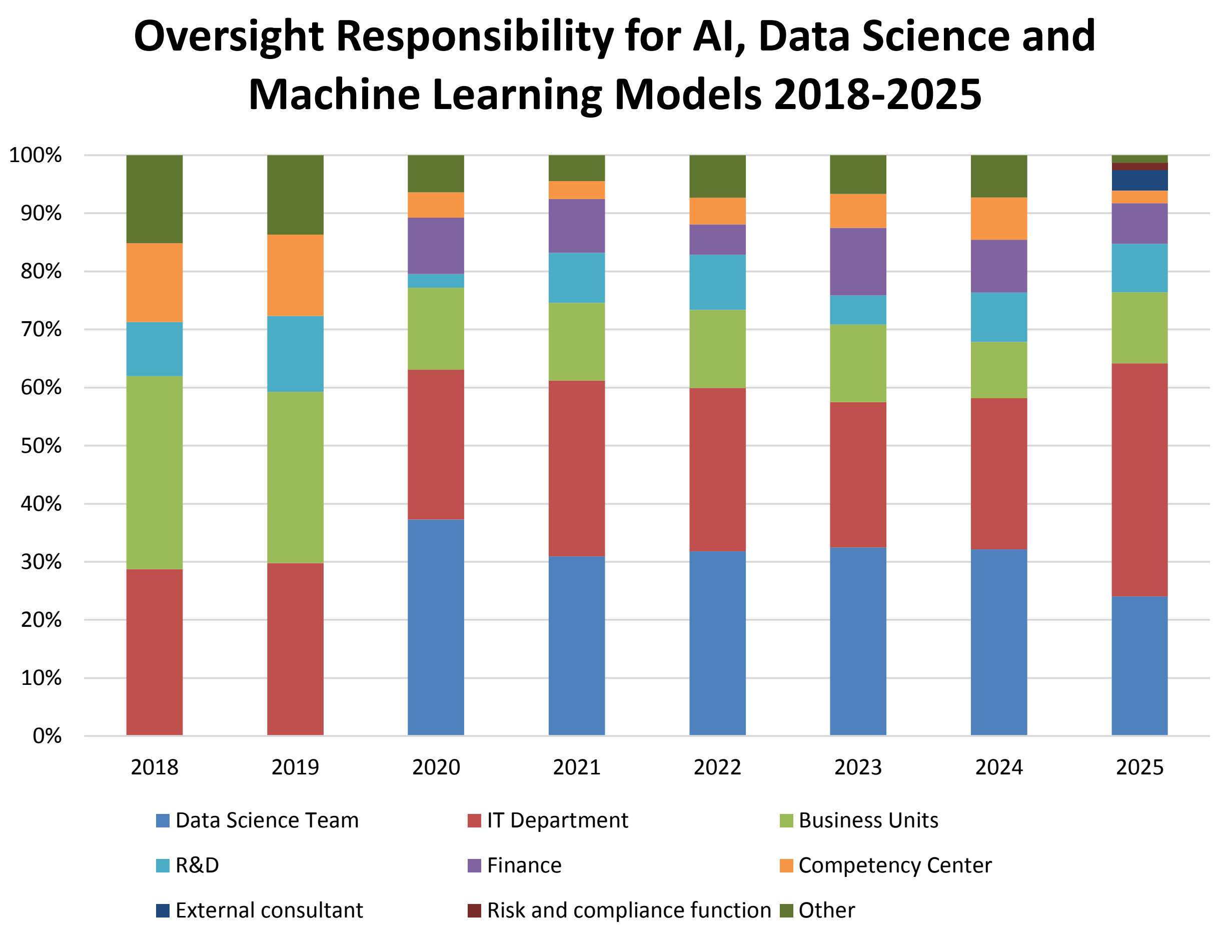


Figure 12 – Oversight responsibility for AI, data science and machine learning models 2018-2025

Geographic variations significantly impact ownership of oversight responsibilities. North America and Asia Pacific both prioritize IT (around 42%), while Latin America shows even stronger IT leadership (54%). Data science teams are more prominent in Asia Pacific (33%) than in North America (20%). Europe, Middle East, and Africa distributes responsibility more broadly, with a notable R&D presence. These regional differences likely reflect varying organizational structures, regulatory landscapes, and levels of AI maturity across different areas.

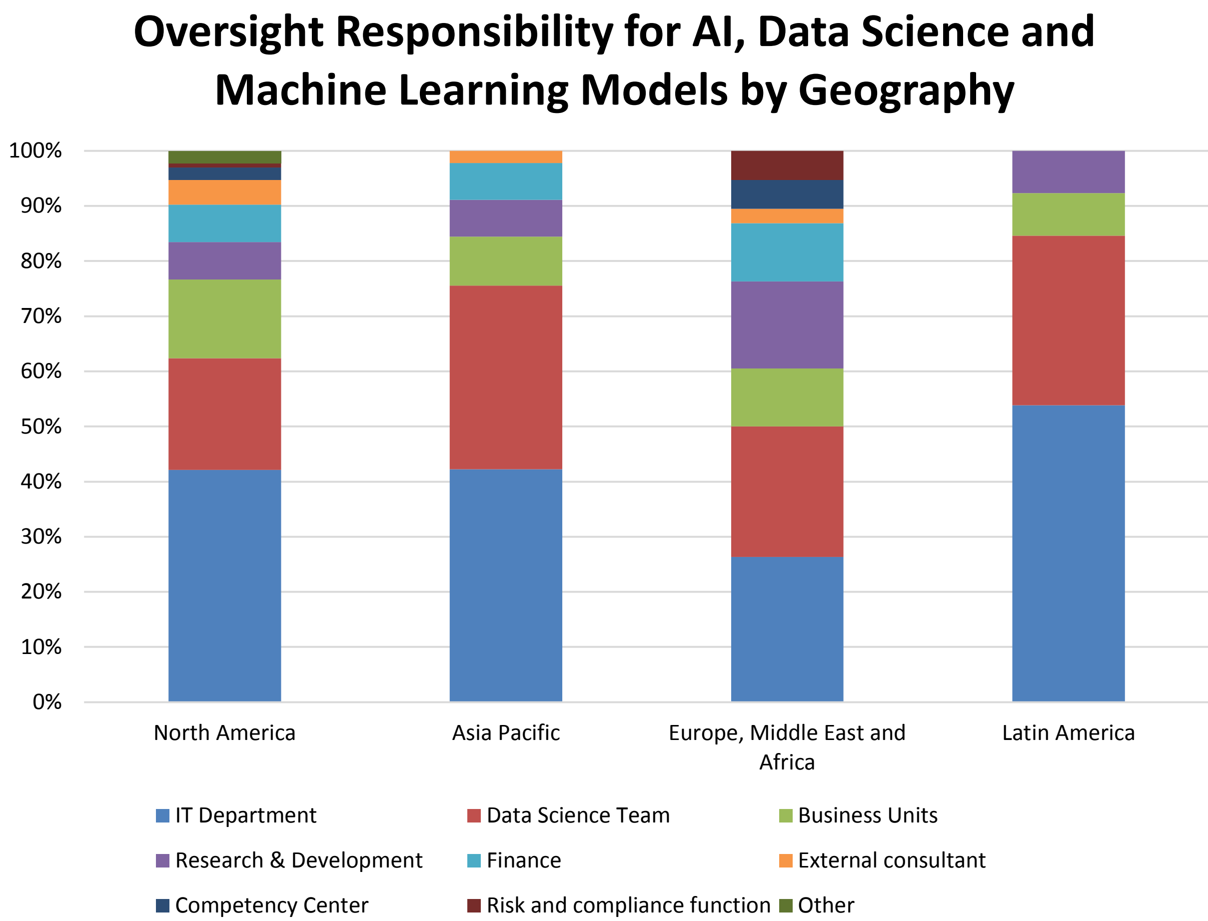


Figure 13 – Oversight responsibility for AI, data science, and machine learning models by geography

AI/ML model oversight responsibility varies by industry, significantly shaping ownership models. Business services and healthcare show strong IT leadership (over 40%), while technology and manufacturing follow closely. Data science teams are most prominent in retail & wholesale (50%), healthcare (44%), and education (24%). Like healthcare, financial services also demonstrates a mixed approach, with both IT and data science having responsibility. These variations likely reflect industry-specific risk profiles, regulatory demands, and levels of AI adoption maturity.

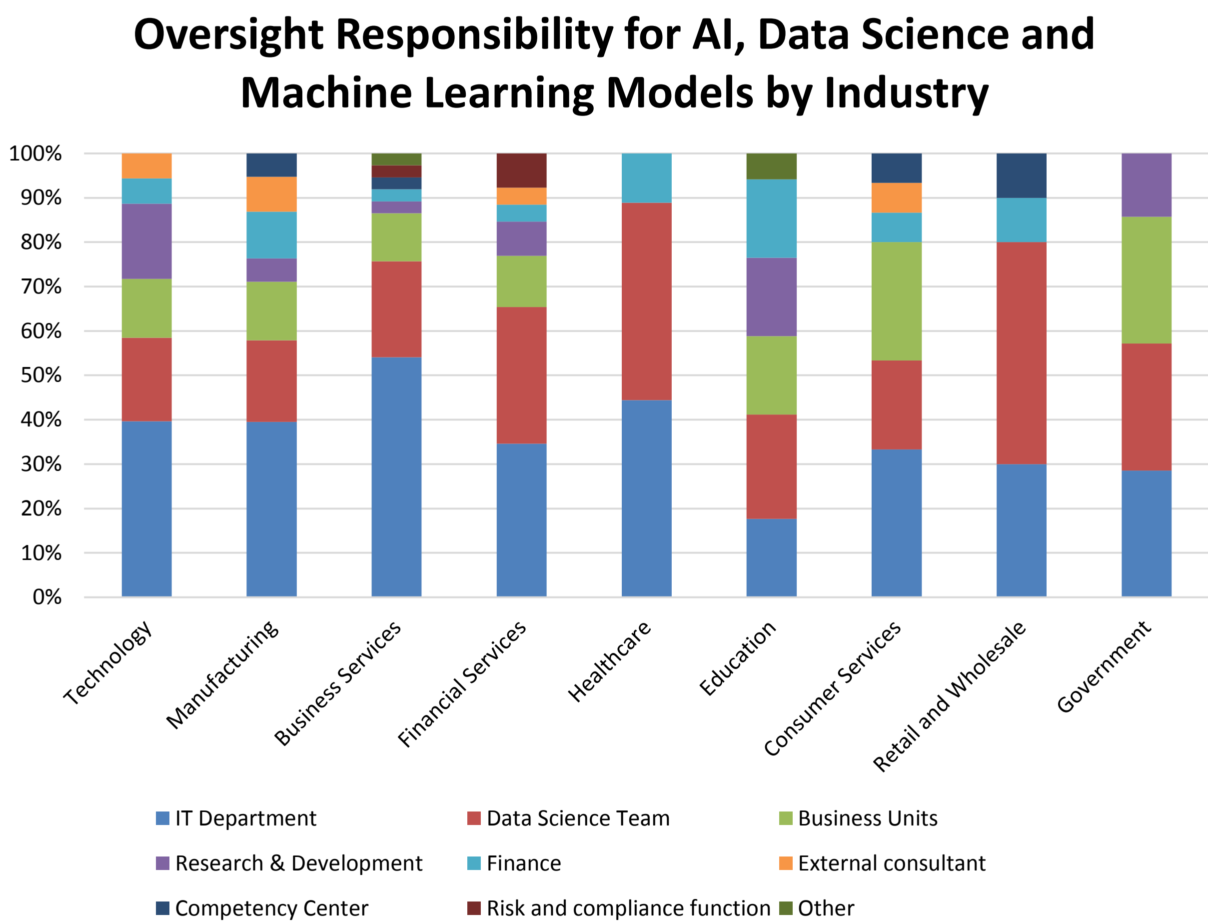


Figure 14 – Oversight responsibility for AI, data science, and machine learning models by industry

Organization size significantly influences ownership and oversight patterns. Notably, very large organizations (more than 10,000 employees) see more business unit involvement than do their smaller counterparts. The data suggests that as companies scale, operational considerations and governance structures become more critical, and responsibility shifts to a mixture of data science teams, IT, and business units. Smaller organizations (1-1,000 employees) show a much broader mix of ownership and oversight for data science and machine learning models.

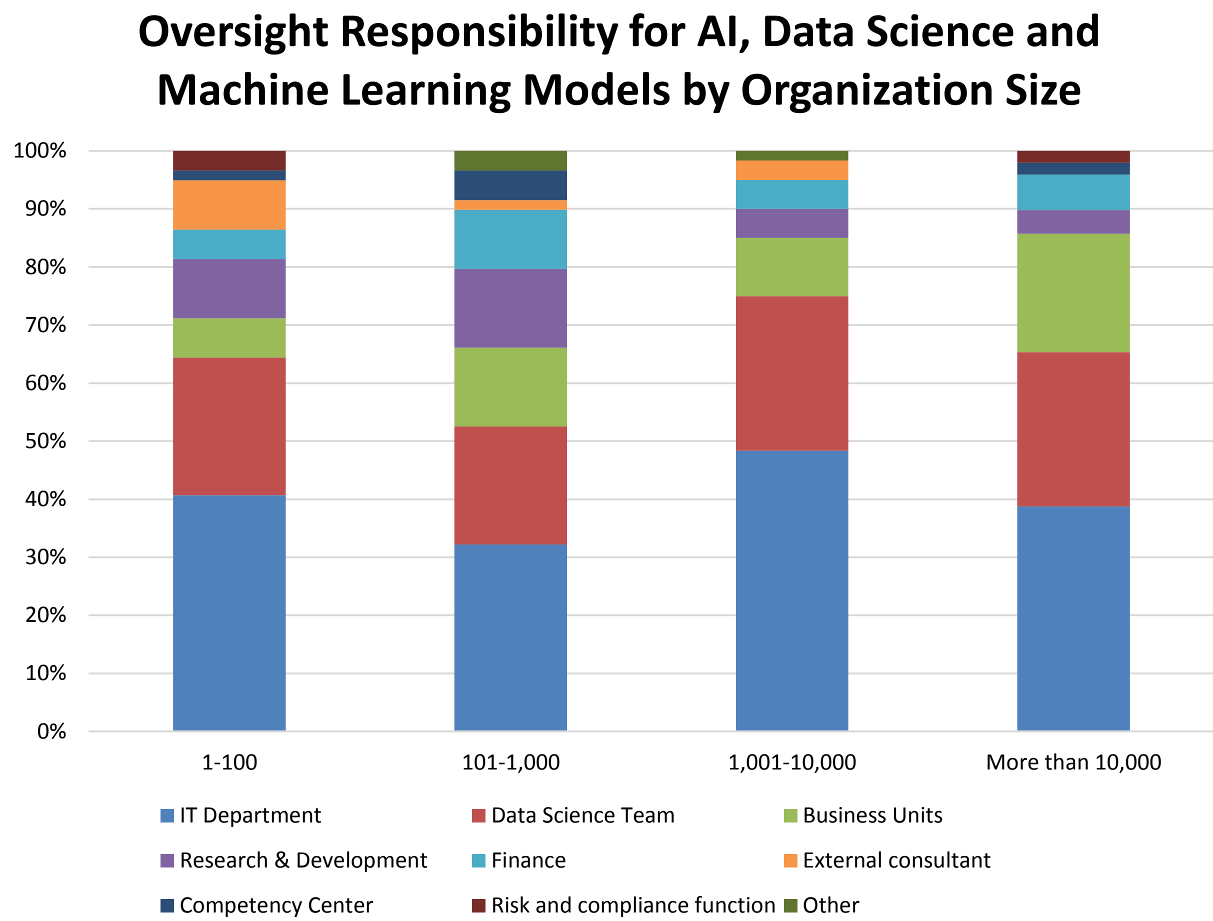


Figure 15 – Oversight responsibility for AI, data science, and machine learning models by organization size

Refresh Frequency for AI, Data Science and Machine Learning Models

In 2025, model refresh patterns diverge from the long‑term baseline: daily updates drop to 18 %, weekly climb to 20%, monthly recede to 25%, quarterly stay steady at 13%, biannually falls below 4%, and annual refreshes increase to 18%. Compared with 2018-2024, the most striking shift is the rebound in annual updates—an almost fourfold jump from the low point of 4% in 2021. This uptick likely reflects the maturation of ModelOps pipelines that now bundle model governance into yearly “audit” cycles, driven by tighter regulatory scrutiny (e.g., GDPR‑style model‑explainability mandates) and the need to certify models before high‑stakes deployments. The modest rise in weekly refreshes suggests continued adoption of automated drift detection, while the decline in monthly updates may indicate that teams are consolidating retraining into fewer, more comprehensive cycles. Overall, the 2025 profile signals a shift toward structured, compliance‑driven refresh regimes rather than ad hoc daily tinkering.

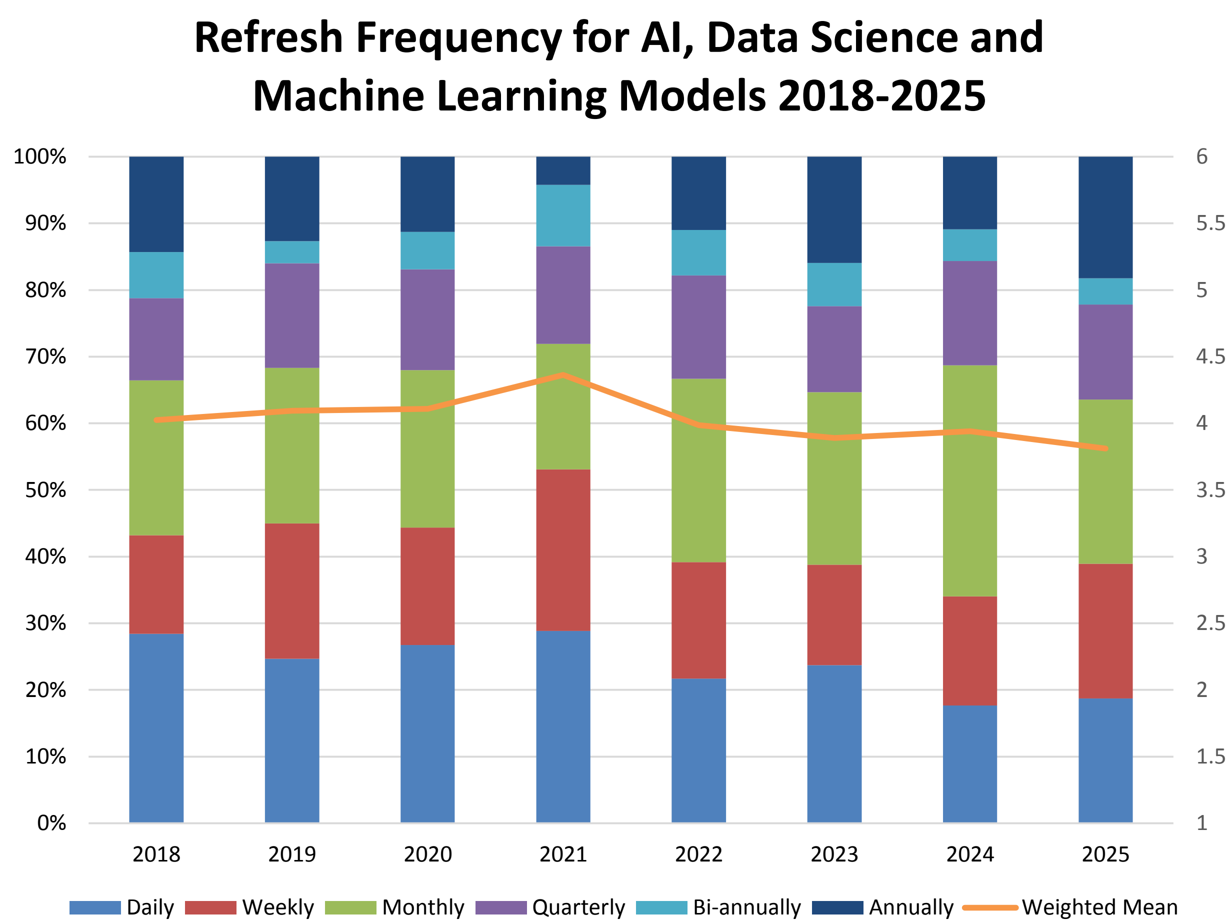


Figure 16 – Refresh frequency for AI, data science, and machine learning models 2018-2025

In 2025, North America leads with a 23% daily update rate—reflecting mature ModelOps and high data velocity—but only 12% refresh annually, indicating continuous retraining has largely replaced yearly audits. Asia Pacific shows the strongest monthly cadence (36%) and a 20% quarterly share. EMEA's profile is unique: 39% of respondents refresh annually, likely driven by GDPR-style regulatory audits; biannual data are absent, suggesting consolidation into quarterly or yearly cycles. Latin America balances a high weekly rate (33%) with a striking 50% annual share. These regional patterns reveal how local regulations, data infrastructure maturity, and industry dynamics shape model refresh strategies.

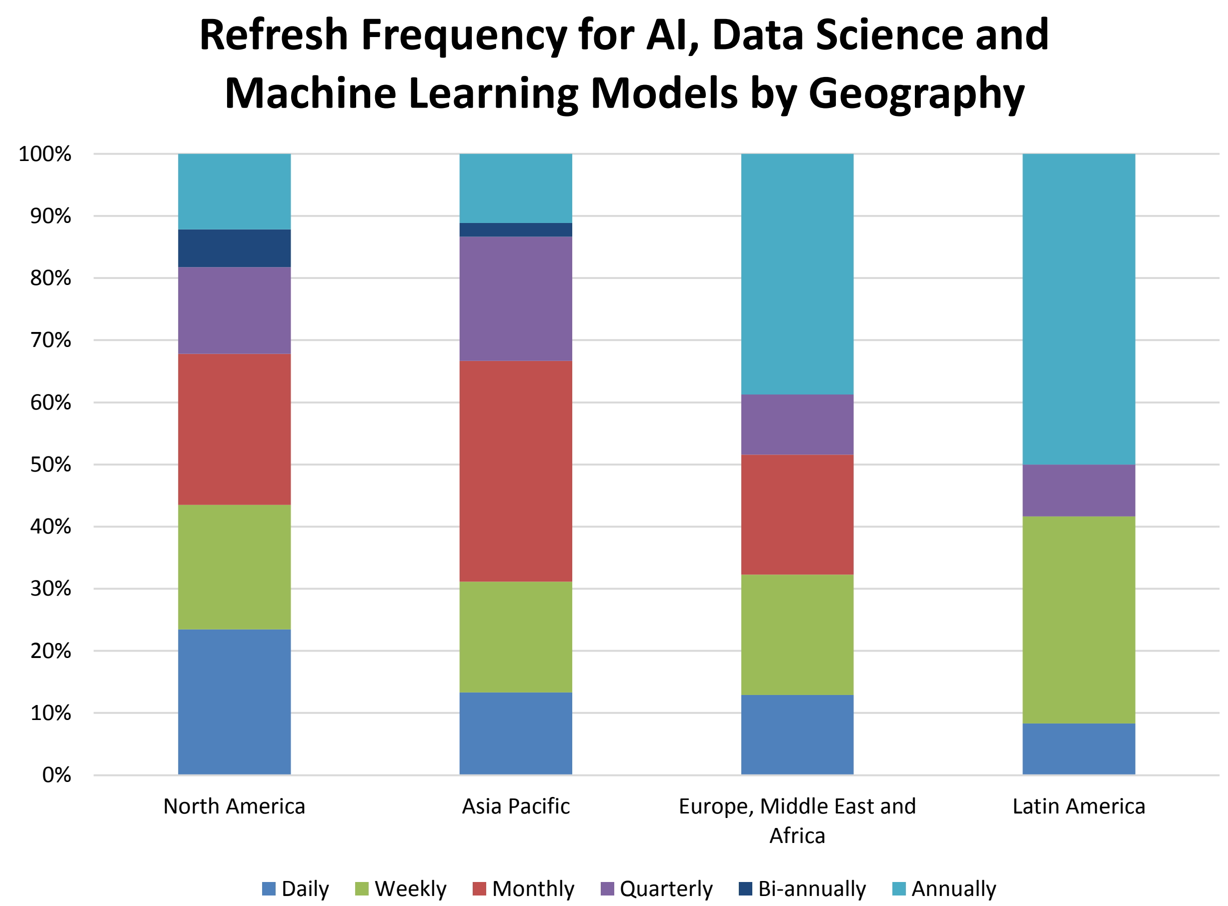


Figure 17 – Refresh frequency for AI, data science, and machine learning models by geography

Sales & marketing leads with daily (25%) and weekly (38%) updates, driven by real-time personalization needs. Finance also favors daily refreshes (38%) but retains a sizable monthly share (24%) for risk models, while regulatory reporting remains quarterly or annual. Operations balances weekly (33%) and monthly (33%) cycles for routine monitoring. IT shows the widest spread—35% monthly, 15% quarterly, and only 3% biannual—reflecting infrastructure upkeep rather than business logic. Executive management prefers daily (19%) and weekly (25%) refreshes for strategic dashboards. BI/analytics spreads across all frequencies with a 26% monthly share, while R&D focuses on monthly (32%) and quarterly (16%) cycles for prototyping. Functions tied to revenue or risk require more frequent updates; infrastructure and experimentation adopt broader intervals.

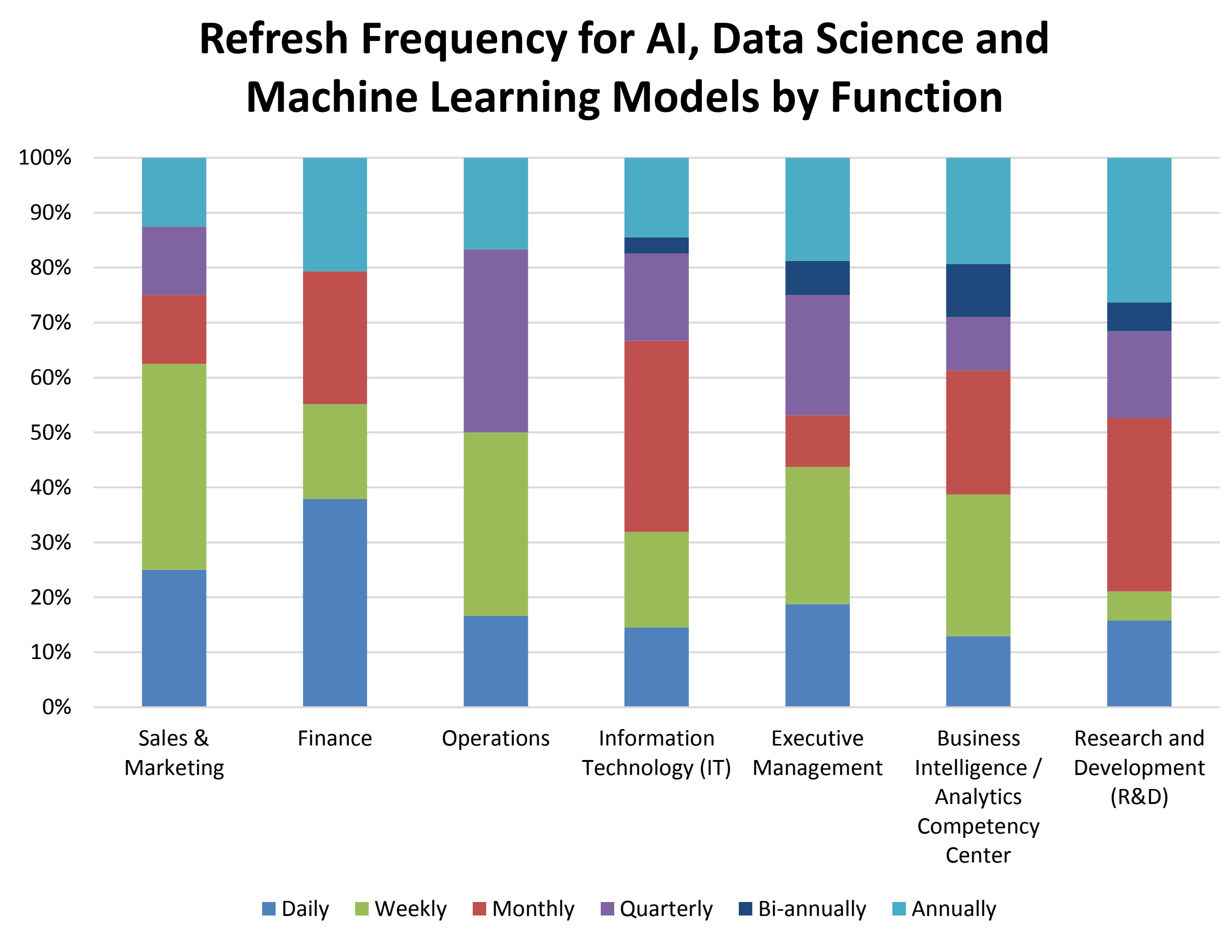


Figure 18 – Refresh frequency for AI, data science, and machine learning models by function

Consumer services dominates daily refreshes (67%), reflecting the need for instant personalization in retail and media, while healthcare spreads evenly across all frequencies—especially biannual (24%) and annual (24%)—indicating regulatory compliance cycles. Manufacturing leans toward monthly updates (27%), with a sizable quarterly share (21%). Technology firms favor monthly refreshes (41%), underscoring data‑driven product iterations, whereas business services mixes daily (25%) and annual (28%) to balance operational efficiency and strategic planning. Financial services distributes evenly across daily to quarterly, but maintains a 25% annual share for audit purposes. Government and retail show uniform distributions with a strong annual tilt (33%–30%), while education's high annual rate (47%) signals long‑term curriculum or policy modeling. Overall, sectors with customer‑facing immediacy prioritize daily updates; regulated or research‑heavy industries favor longer cycles.

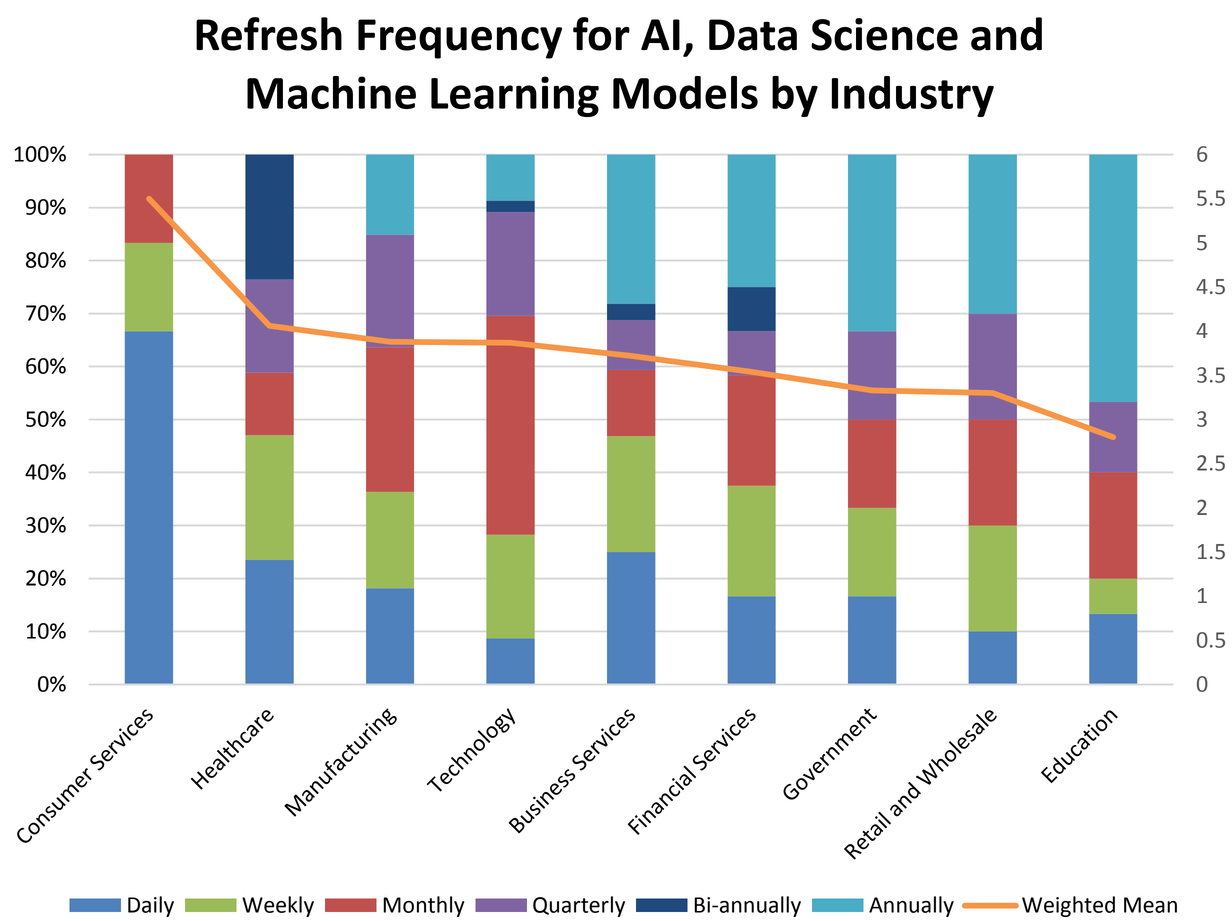


Figure 19 – Refresh frequency for AI, data science, and machine learning models by industry

Small firms (1–100 employees) allocate most effort to monthly refreshes (25%), with a modest weekly share (16%). Midsize companies (101–1,000) mirror this pattern but add a larger quarterly component (20%) and drop biannual use to less than 4%. As organizations grow beyond 1,000 employees, daily updates become dominant—21% for 1,001–10,000 and 26% for more than 10,000—while weekly refreshes rise sharply (26% in the 1,001–10,000 bracket). Monthly refreshes decline to 15% for midsize firms but surge again to 36% for the largest enterprises, indicating a hybrid strategy that balances real‑time needs with batch processing. Overall, larger organizations lean toward more frequent daily updates yet still maintain substantial monthly cycles.

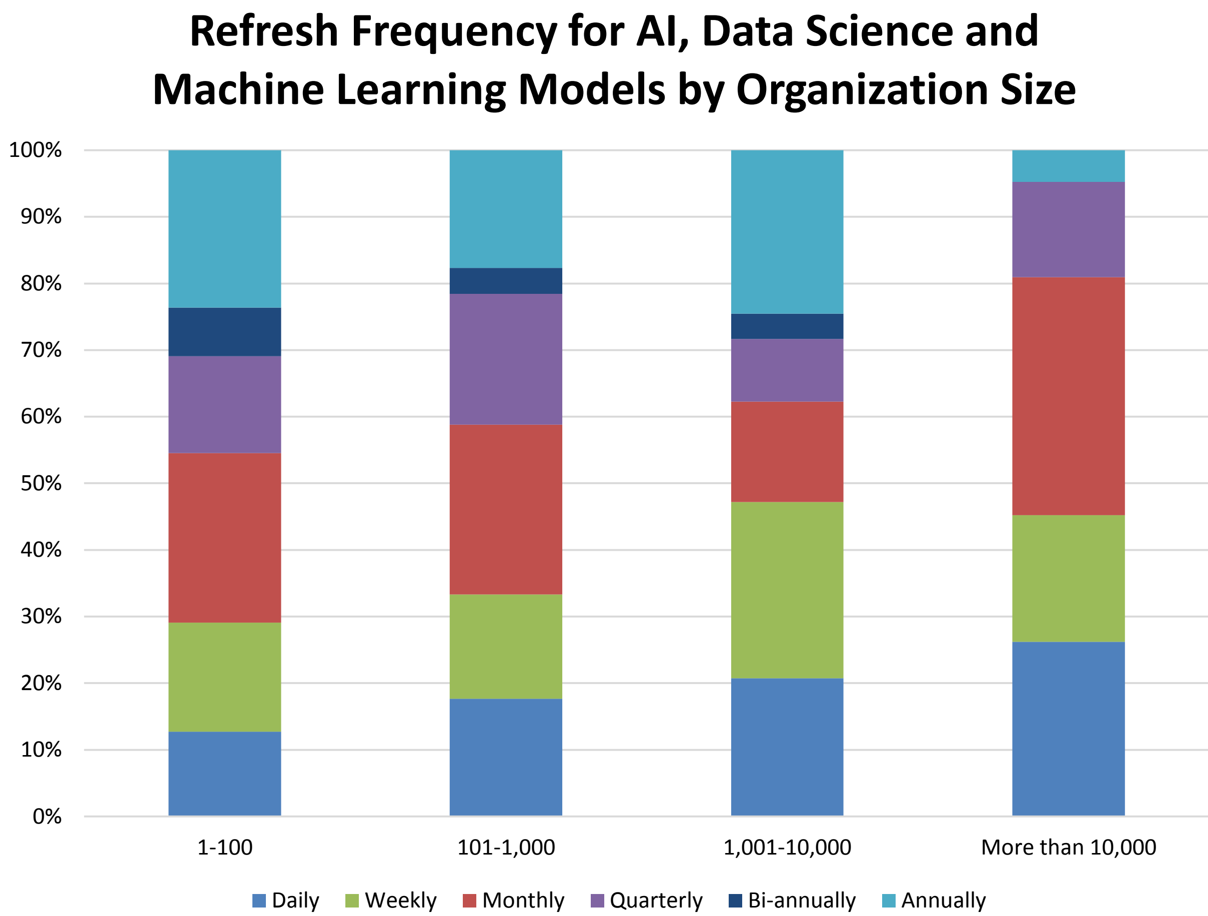


Figure 20 – Refresh frequency for AI, data science, and machine learning models by organization size

For organizations that have agentic AI workflows in production, the model refresh schedule is relatively conservative: 25% of updates occur weekly or monthly, while only 13.9% are daily, and a modest 22% happen quarterly. This reflects a focus on stability and risk mitigation in live deployments. By contrast, teams experimenting with agentic AI push for more frequent iteration—20% cite daily refreshes—and they also schedule a surprisingly high annual cadence (22%) to capture long‑term learning signals. For these teams, weekly updates drop to 19%, and quarterly refreshes fall below 12%. The experimental cohort's heavier reliance on daily and yearly cycles indicates an emphasis on rapid prototyping, continuous learning, and periodic model retraining to validate emergent behaviors.

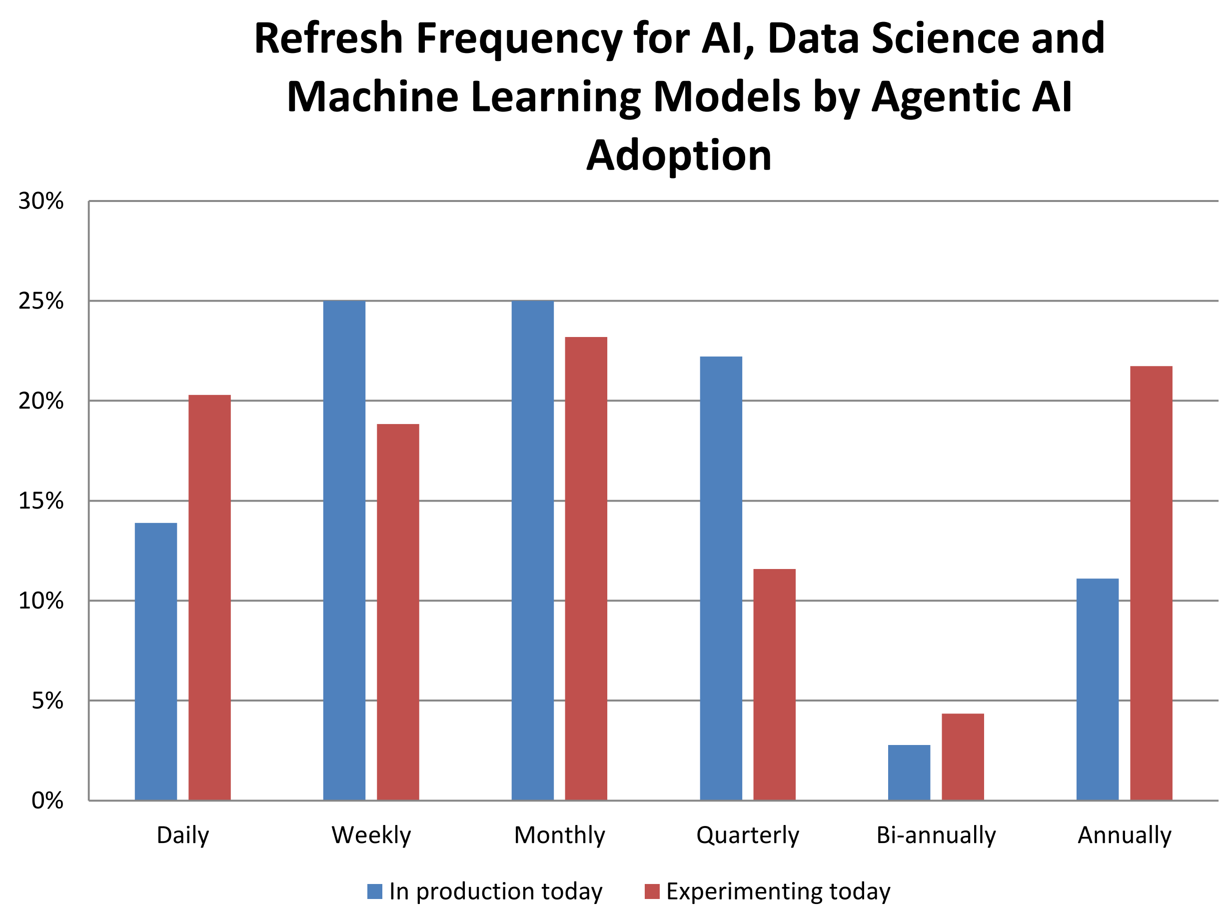


Figure 21 – Refresh frequency for AI, data science, and machine learning models by agentic AI adoption

Model Operations Feature Priorities

We asked respondents which ModelOps features are important for AI, data science, and machine learning models. They tell us the importance of ModelOps capabilities varies, depending on the organization and the stage of their projects. The highest combined scores for the critical and very important ratings cluster around end‑to‑end model governance: life cycle management (45%), lineage/history (43%), and version control (42%). These indicate that teams prioritize traceability, reproducibility, and the ability to roll back or retire models. Monitoring & alerting (41%-42 %) and continuous integration/continuous deliverability/continuous testing (40%) follow closely, underscoring a need for automated observability and deployment pipelines. Searchable repositories (40%) and collaborative tools (39%) also rank high, reflecting a shift toward shared model libraries and co‑creation. Lower scores on SageMaker integration (24%) suggest that while cloud tools matter, the community places higher value on broader, platform‑agnostic capabilities.

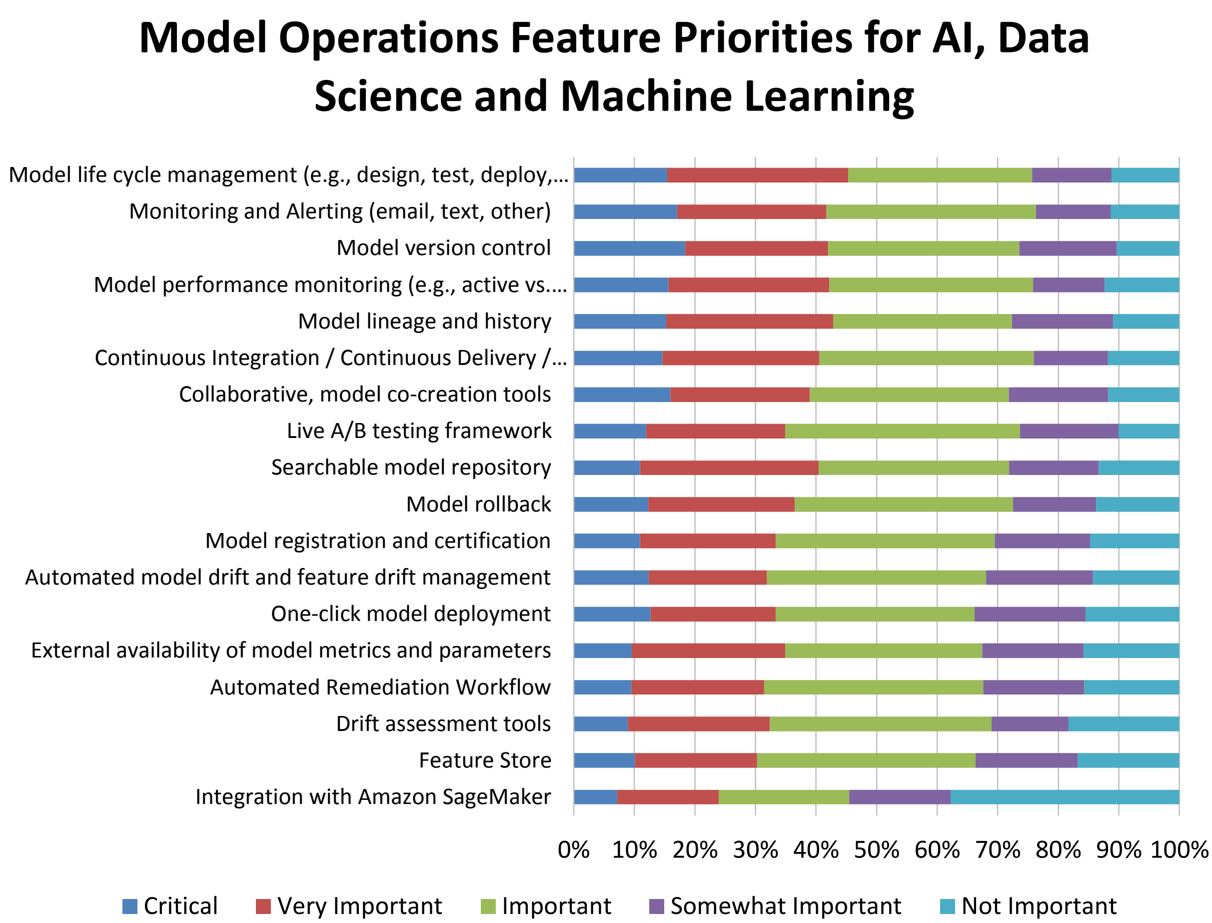


Figure 22 – Model operations feature priorities for AI, data science, and machine learning

Across regions, the most valued ModelOps capabilities cluster on collaboration, governance, and monitoring. Asia Pacific leads with collaborative tools (3.64), a searchable repository (3.61), lineage/history (3.60), and version control (3.62), reflecting mature traceability practices. Europe, the Middle East and Africa prioritize continuous integration/continuous delivery (CI/CD; 3.23), life cycle management (3.05), monitoring/alerting (3.08), and performance monitoring (3.16). North America mirrors this mix, emphasizing life‑cycle, CI/CD, monitoring, and performance. Latin America balances collaborative tools (3.08) and CI/CD (3.17) with lower version control (3.00) and lineage (3.00). These patterns suggest that while ModelOps engineering rigor is universal, in many regions governance tooling remains an unmet need and growth area. (Note: The survey used a scale of 1‑5, in which scores of 1 are considered noncritical, whereas 5 denotes critical importance for ModelOps success.)

Model Operations Feature Priorities for AI, Data Science and Machine Learning by Geography

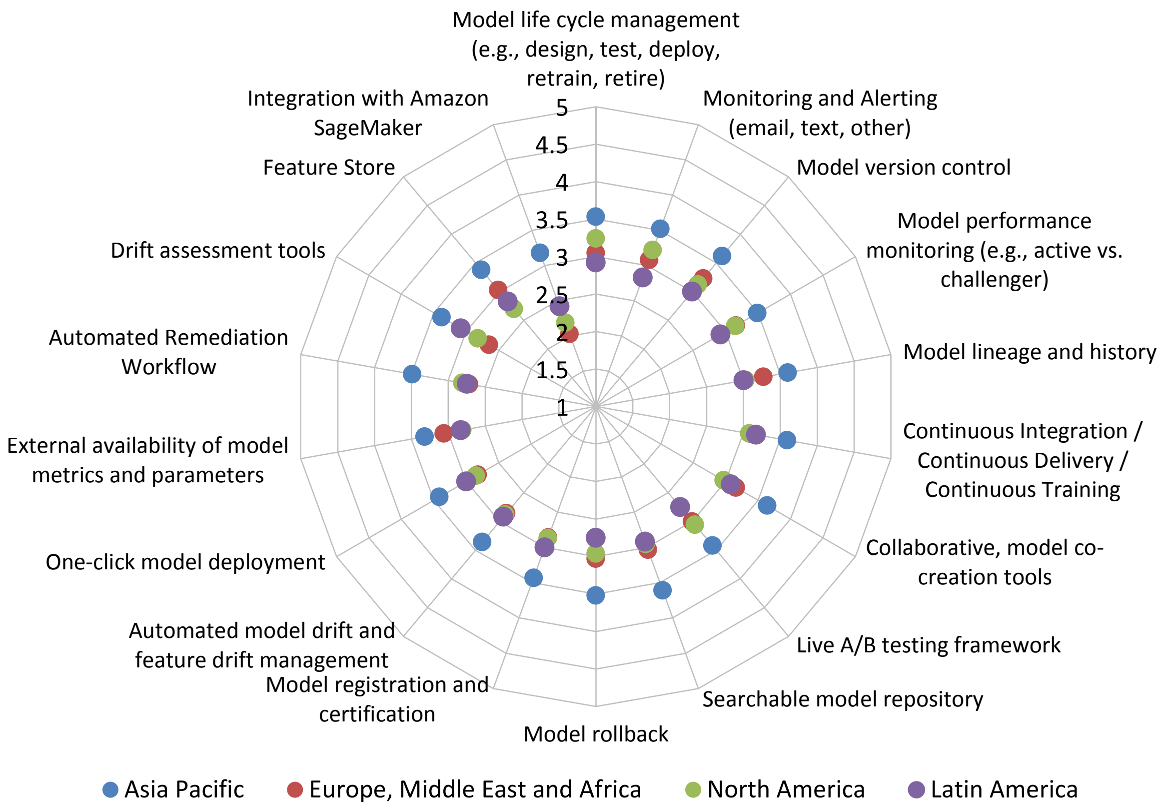


Figure 23 – Model operations feature priorities for AI, data science, and machine learning by geography

The chart shows that business intelligence/analytics and IT consistently rate most capabilities above 3, underscoring their heavy reliance on robust model management pipelines. Sales & marketing places the highest emphasis on monitoring (3.56) and lineage (3.67), reflecting a need for real‑time visibility into model performance that directly impacts customer engagement. Finance scores lower overall, with low ratings of automated remediation (2.66) and feature store usage (2.22), suggesting tighter risk controls but less automation. R&D values version control (2.95) and drift assessment (2.53) moderately, while executive management shows the weakest focus across all items, indicating that strategic oversight remains high‑level rather than operational. These patterns highlight where investment in tools will yield the greatest return for each function.

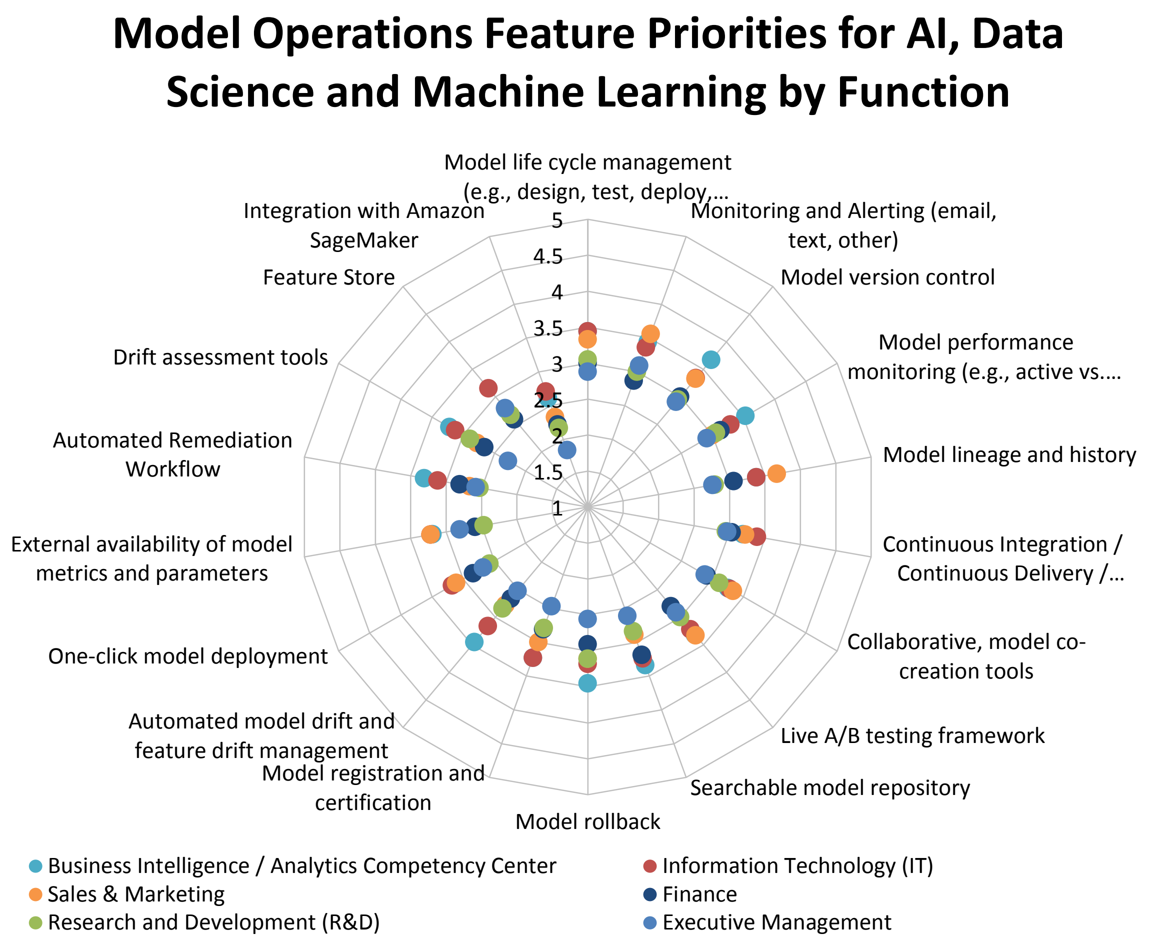


Figure 24 – Model operations feature priorities for AI, data science, and machine learning by function

Across industries, technology and manufacturing consistently rate the highest on core capabilities such as model life cycle management (3.5-3.44) and CI/CD/CT (around 3.53), reflecting mature ModelOps pipelines. Consumer services shows strong emphasis on monitoring & alerting (3.42) and performance monitoring (3.62), indicating a focus on real-time model health for customer-facing applications. Financial services prioritizes version control (3.46) but scores lower on drift management (2.92) and external metrics (2.84), suggesting tighter governance but less automation. Healthcare values lineage (3.06) and collaborative tools (3.19) while lagging in feature store adoption (2.82). Government stands out with the highest monitoring score (3.71) yet low feature-store use (1.71), highlighting a need for better data infrastructure. Overall, ModelOps maturity varies markedly by sector, guiding where tooling investments will yield the greatest impact.

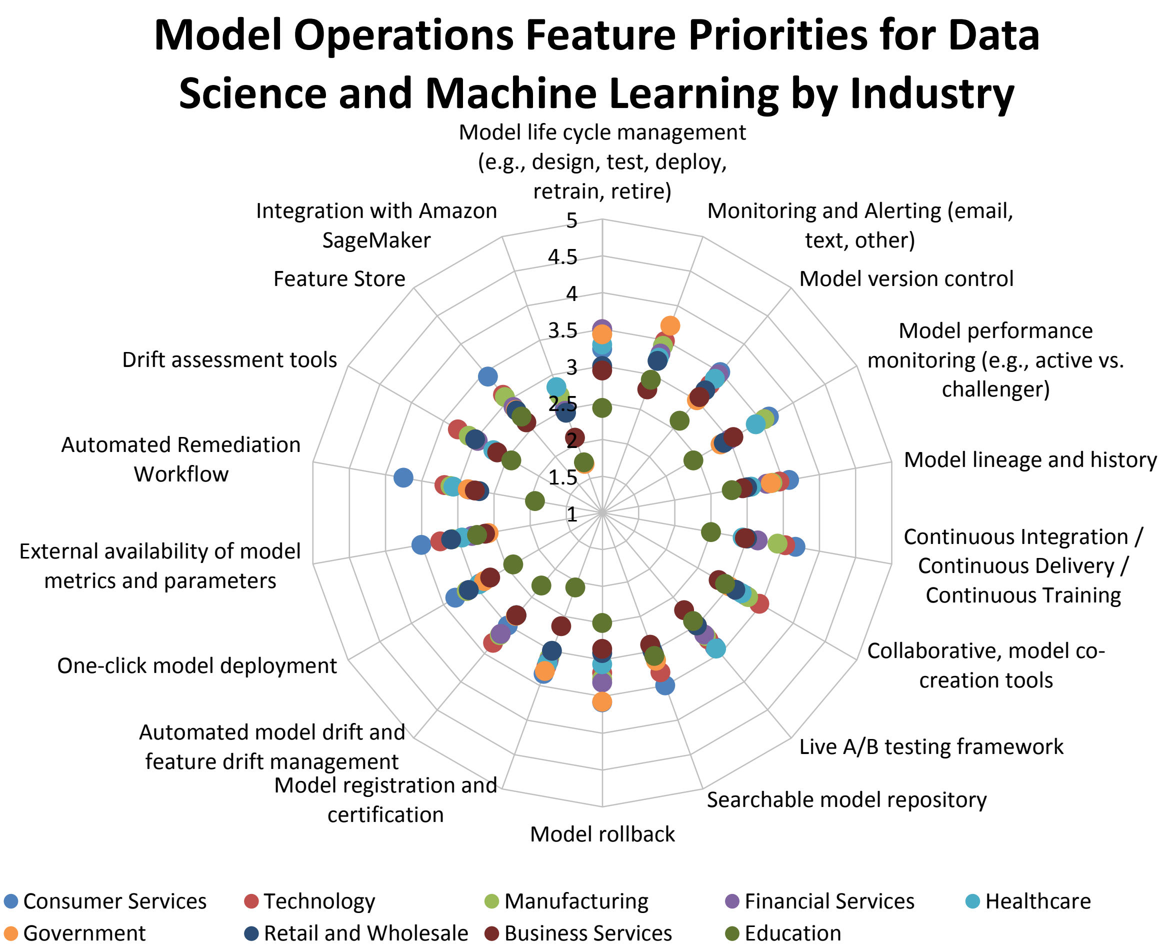


Figure 25 – Model operations feature priorities for AI, data science, and machine learning by industry

The data reveal a clear “size‑scale” effect: Larger organizations (more than 10, 000 employees) consistently rate every capability higher, with the most pronounced gaps in model life‑cycle management (3.79 vs. 2.98) and performance monitoring (3.71 vs. 2.97). Midsize firms (1,001–10,000 employees) show strong emphasis on version control (3.35) and lineage (3.35), suggesting mature governance but less focus on rapid deployment tools. Small companies (100 or fewer employees) prioritize collaborative tools (3.07) over automation, reflecting limited resources for CI/CD or drift detection. Feature store adoption remains low across all organization sizes, especially in the smallest group (2.19), indicating a widespread opportunity to improve data reuse and feature consistency. Overall, scaling up in organization size correlates with higher investment in end‑to‑end ModelOps capabilities.

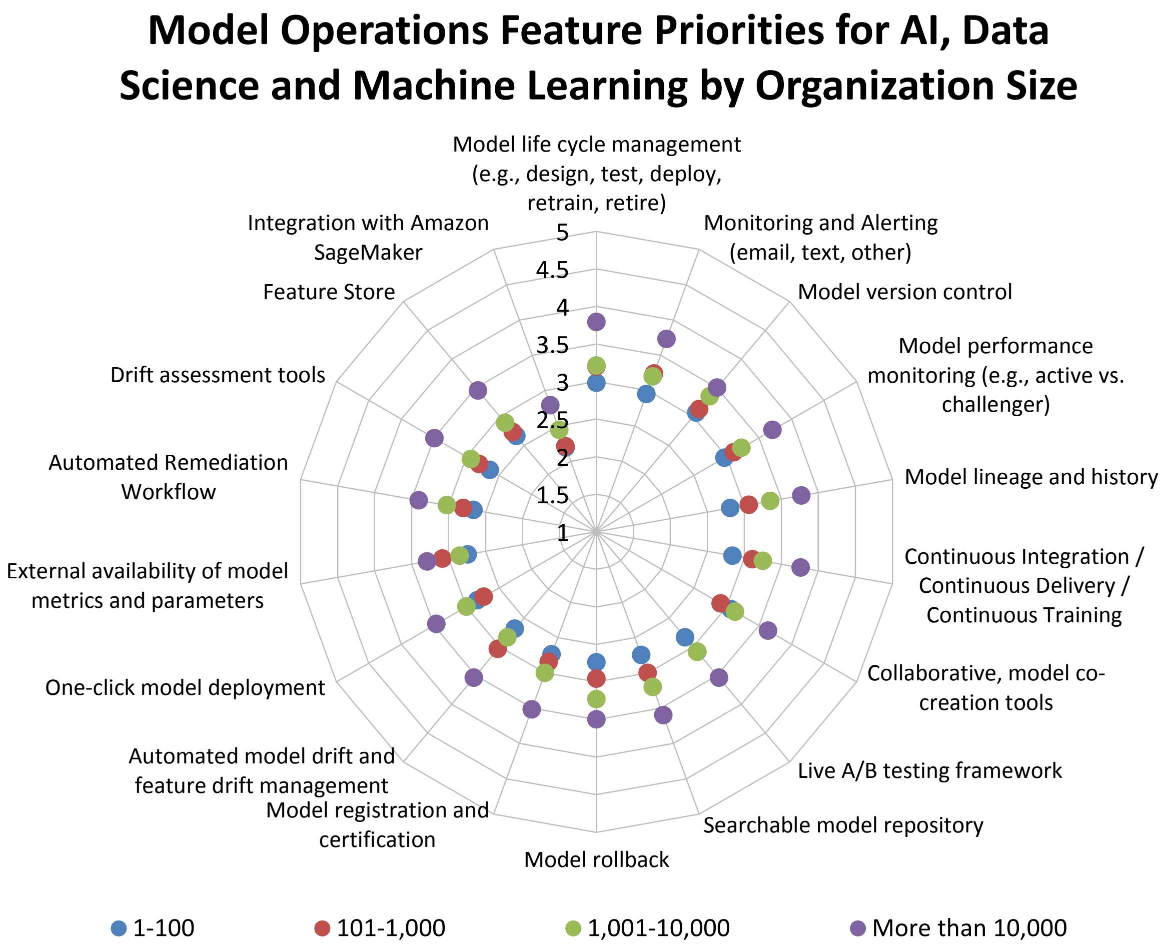


Figure 26 – Model operations feature priorities for AI, data science, and machine learning by organization size

Industry and Vendor Analysis

Industry and Vendor Analysis

Today's vendors overwhelmingly support foundational ModelOps functions—model version control (62%), one‑click deployment (65%), and monitoring/alerting (59%). Over the next 12-24 months, only a handful of plans surface across vendors: live A/B testing, automated remediation, and feature store support each figure in the plans of less than 20% of respondents, while most vendors report "no plans" for these features (about 50% for A/B testing, 54% for remediation). Continuous integration/delivery remains the strongest current offering (52%) but drops sharply in future roadmaps. Feature‑store and drift‑management capabilities lag, with only 35% supporting these today and less than 16 % projected. Overall, while vendors are solidifying core infrastructure, many advanced capabilities—especially those that enable experimentation and automated governance—remain largely unplanned or are in the early stages.

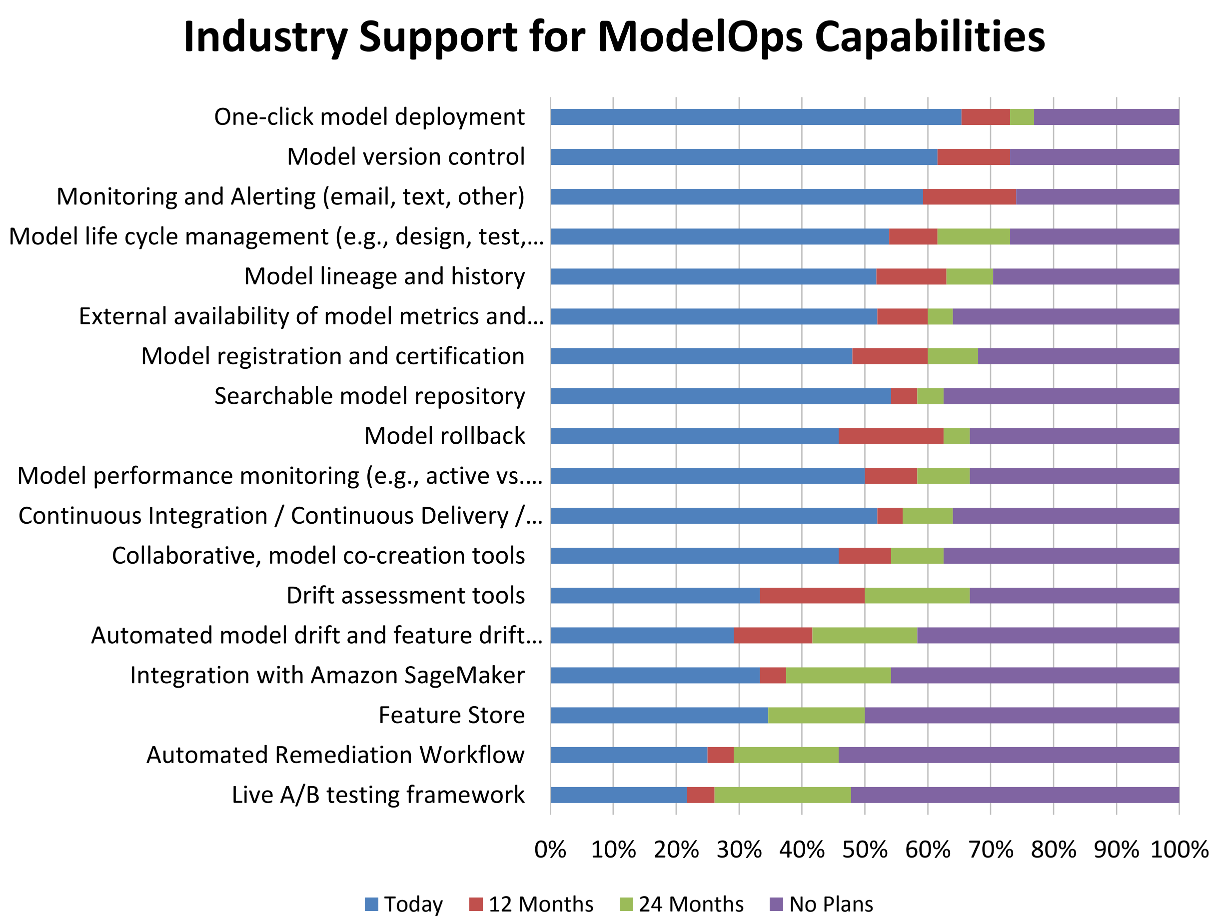


Figure 27 – Industry support for ModelOps capabilities

ModelOps Vendor Ratings

In rating the vendors, we consider life-cycle management, development, governance, operations, and monitoring (fig. 28). It is important to scrutinize all rating categories and match vendor strengths to use cases and requirements. Ratings take into consideration vendor provided information, which is confirmed and then validated with user input.

The top half of ranked vendors are in a tight grouping with multiple ties. For first place there is a two-way tie with Domino Data Lab, and Palantir. Alteryx is in second place. Third-place vendors are Amazon AWS, Domo, and KNIME. Altair, Dataiku, and Qlik are tied for fourth place. Fifth place vendors are DataRobot, Oracle, Snowflake and Tableau.

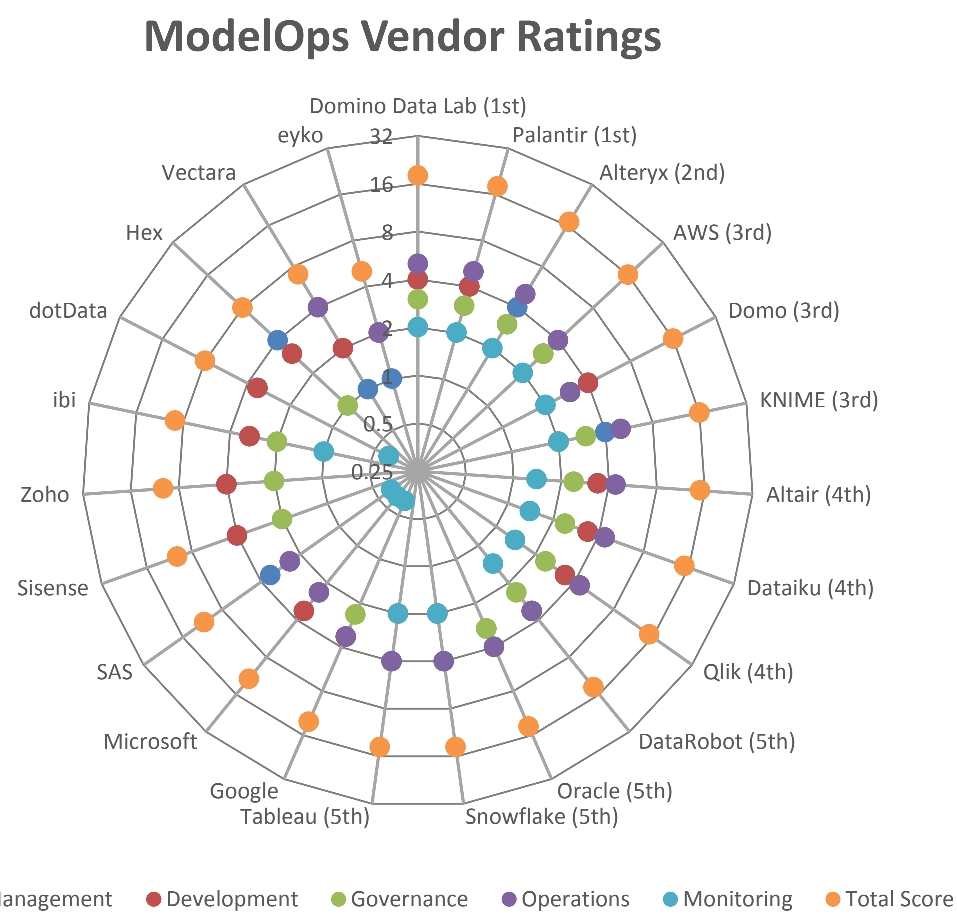


Figure 28 – ModelOps vendor ratings

*A logarithmic scale is used for the scoring chart to address skewness towards larger values

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Appendix: ModelOps User Survey Instrument

Please enter your contact information below

First Name*: ______________________________

Last Name*: ______________________________

Title: ______________________________

Company Name*: ______________________________

Street Address: ______________________________

City: ______________________________

State: ______________________________

Zip: ______________________________

Country: ______________________________

Email Address*: ______________________________

Phone Number: ______________________________

URL: ______________________________

What major geography do you reside in?*

( ) North America

( ) Europe, Middle East and Africa

( ) Latin America

( ) Asia Pacific

Please identify your primary industry*

( ) Advertising

( ) Aerospace

( ) Agriculture

( ) Apparel & Accessories

( ) Automotive

( ) Aviation

( ) Biotechnology

( ) Broadcasting

( ) Business Services

( ) Chemical

( ) Construction

( ) Consulting

( ) Consumer Products

( ) Defense

( ) Distribution & Logistics

( ) Education (Higher Ed)

( ) Education (K-12)

( ) Energy

( ) Entertainment and Leisure

( ) Executive search

( ) Federal Government

( ) Financial Services

( ) Food, Beverage and Tobacco

( ) Healthcare

( ) Hospitality

( ) Insurance

( ) Legal

( ) Manufacturing

( ) Mining

( ) Motion Picture and Video

( ) Not for Profit

( ) Pharmaceuticals

( ) Publishing

( ) Real estate

( ) Retail and Wholesale

( ) Sports

( ) State and Local Government

( ) Technology

( ) Telecommunications

( ) Transportation

( ) Utilities

( ) Other - Please specify below

How many employees does your company employ worldwide?

( ) 1-100

( ) 101-1,000

( ) 1,001-2,000

( ) 2,001-5,000

( ) 5,001-10,000

( ) More than 10,000

What function do you report into?*

( ) Business Intelligence Competency Center

( ) Executive Management

( ) Finance

( ) Human Resources

( ) Information Technology (IT)

( ) Marketing

( ) Operations (e.g., Manufacturing, Supply Chain, Services)

( ) Research and Development (R&D)

( ) Sales

( ) Strategic Planning Function

( ) Other - Write In

How many predictive models are currently in production?

( ) Don't know

( ) 1-25 models

( ) 26-50 models

( ) 51-100 models

( ) 101-250 models

( ) More than 250 models

How often are predictive models refreshed?

( ) Daily

( ) Weekly

( ) Monthly

( ) Quarterly

( ) Bi-annually

( ) Annually

145) Who is responsible for monitoring/managing ML / predictive models (ModelOps)?

( ) IT Department

( ) Business Units

( ) Finance

( ) Competency Center

( ) Data Science Team

( ) Research & Development

( ) Other - Write In: ____

Which of the following model operations (ModelOps) features are important for data science and machine learning?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**
Model life cycle management (e.g., design, test, deploy, retrain, retire)	( )	( )	( )	( )	( )
Model registration	( )	( )	( )	( )	( )

and certification					
Model version control	( )	( )	( )	( )	( )
Model lineage and history	( )	( )	( )	( )	( )
Model rollback	( )	( )	( )	( )	( )
One-click model deployment	( )	( )	( )	( )	( )
Searchable model repository	( )	( )	( )	( )	( )
Collaborative, model co-creation tools	( )	( )	( )	( )	( )
Model performance monitoring (e.g., active vs. challenger)	( )	( )	( )	( )	( )
Drift assessment tools	( )	( )	( )	( )	( )
Model explainability	( )	( )	( )	( )	( )
Live A/B testing framework	( )	( )	( )	( )	( )

Feature store	( )	( )	( )	( )	( )